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1 **Medical Image Segmentation in a Multiple**
2 **Labelers Context: Application to the Study**
3 **of Histopathology**

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Medical Image Segmentation in a Multiple Labelers Context: Application to the Study of Histopathology

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27 **Segmentación de Imágenes Médicas en un**
28 **Contexto de Múltiples Anotadores:**
29 **Aplicación al Estudio de Histopatologías**

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56 meaningfully to the field of early cancer detection.

ABSTRACT

60 Breast cancer remains one of the deadliest diseases affecting women worldwide,
61 making accurate diagnosis and treatment planning crucial for improving patient
62 outcomes. Semantic segmentation of histopathology images plays a vital role in
63 this process, as it enables precise identification and analysis of tissue structures.
64 However, the current approaches face significant challenges in real-world
65 scenarios where multiple annotators with varying expertise and reliability
66 contribute to the labeling process. This crowdsourcing paradigm, while
67 cost-effective, introduces inconsistencies that can compromise the quality of the
68 segmentation results.

69 This thesis addresses two fundamental challenges in medical image segmentation:
70 the inherent uncertainty in multiple annotator scenarios and the need for robust
71 segmentation models that can handle varying image quality and annotator
72 expertise. These challenges arise from the complex nature of histopathology
73 images, where tissue structures can be ambiguous, and the varying levels of
74 expertise among annotators lead to inconsistent labeling patterns. Additionally,
75 the high cost and time requirements for expert annotations create a need for
76 efficient solutions that can work with crowdsourced data.

77 To address these challenges, this thesis proposes a comprehensive approach that
78 combines novel loss functions with robust deep learning architectures to handle
79 multiple annotator scenarios effectively.

80 The first objective introduces a novel loss function that implicitly infers optimal
81 segmentation while assessing labeler reliability without requiring explicit
82 performance inputs. This approach differs from conventional methods by
83 eliminating the need for direct supervision of labeler performance, instead using
84 an intermediate reliability map that enables the model to prioritize information
85 from reliable labelers while disregarding noisy labels. The mathematical
86 foundation of this approach lies in the probabilistic modeling of annotator
87 reliability across the input space.

88 The second objective focuses on the development of a robust segmentation
89 architecture that combines the strengths of U-shaped deep learning models with
90 the proposed reliability-aware loss function. This combination enables the model
91 to learn both the segmentation task and the reliability patterns simultaneously,
92 leading to more accurate and consistent results. The architecture's key innovation
93 lies in its ability to adapt to varying image quality and annotator expertise levels
94 without requiring explicit performance metrics.

95 The third objective involves validating the proposed approach through extensive
96 experimentation on both synthetic datasets and real-world histopathology images.
97 The evaluation demonstrates superior performance compared to state-of-the-art
98 approaches in terms of segmentation accuracy and uncertainty quantification. The
99 results show that the proposed method can effectively handle inconsistent
100 annotations while maintaining high segmentation accuracy, making it particularly
101 valuable for clinical applications.

102 The research makes significant contributions to the field by providing a robust
103 solution for handling inconsistent annotations in medical image segmentation,
104 particularly valuable in histopathology where expert annotations are expensive
105 and time-consuming to obtain. The proposed approach shows promising results in
106 reducing annotation costs while maintaining high segmentation accuracy, making
107 it particularly relevant for clinical applications where precise segmentation of
108 tissue structures is crucial for diagnosis and treatment planning. This work opens

109 new research directions in the areas of crowdsourced medical image analysis,
110 uncertainty quantification in deep learning, and the development of more efficient
111 annotation protocols for medical imaging.

112 **Keywords:** Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer,
113 Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image
114 Analysis

116 El cáncer de mama sigue siendo una de las enfermedades más mortales en la
117 población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la
118 planificación del tratamiento sean cruciales para mejorar las condiciones de los
119 pacientes. La segmentación semántica de imágenes histopatológicas juega un
120 papel vital en este proceso, ya que permite la identificación y análisis preciso de
121 estructuras tisulares. Sin embargo, los enfoques actuales enfrentan desafíos
122 significativos en escenarios del mundo real donde múltiples anotadores con
123 diferentes niveles de experiencia y confiabilidad contribuyen al proceso de
124 etiquetado. Este paradigma de crowdsourcing, aunque rentable, introduce
125 inconsistencias que pueden comprometer la calidad de los resultados de
126 segmentación.

127 Esta tesis aborda dos desafíos fundamentales en la segmentación de imágenes
128 médicas: la incertidumbre inherente en escenarios de múltiples anotadores y la
129 necesidad de modelos de segmentación robustos que puedan manejar la calidad
130 variable de las imágenes y la experiencia de los anotadores. Estos desafíos surgen
131 de la naturaleza compleja de las imágenes histopatológicas, donde las estructuras
132 tisulares pueden ser ambiguas, y los diferentes niveles de experiencia entre los
133 anotadores conducen a patrones de etiquetado inconsistentes. Además, el alto
134 costo y los requisitos de tiempo para las anotaciones de expertos crean la
135 necesidad de soluciones eficientes que puedan trabajar con datos crowdsourced.

El primer objetivo introduce una función de pérdida novedosa que infiere implícitamente la segmentación óptima mientras evalúa la confiabilidad de los etiquetadores sin requerir entradas explícitas sobre su rendimiento. Este enfoque difiere de los métodos convencionales al eliminar la necesidad de supervisión directa del rendimiento de los etiquetadores, utilizando en su lugar un mapa de confiabilidad intermedio que permite al modelo priorizar la información de etiquetadores confiables mientras descarta etiquetas ruidosas. La base matemática de este enfoque radica en el modelado probabilístico de la confiabilidad de los anotadores a través del espacio de entrada.

El segundo objetivo se centra en el desarrollo de una arquitectura de segmentación robusta que combina las fortalezas de los modelos de aprendizaje profundo en forma de U con la función de pérdida propuesta que tiene en cuenta la confiabilidad. Esta combinación permite que el modelo aprenda tanto la tarea de segmentación como los patrones de confiabilidad simultáneamente, lo que lleva a resultados más precisos y consistentes. La innovación clave de la arquitectura radica en su capacidad para adaptarse a diferentes niveles de calidad de imagen y experiencia de los anotadores sin requerir métricas de rendimiento explícitas.

El tercer objetivo valida el enfoque propuesto a través de experimentación exhaustiva tanto en conjuntos de datos sintéticos como en imágenes histopatológicas del mundo real. La evaluación demuestra un rendimiento superior en comparación con los enfoques más avanzados en términos de precisión de segmentación y cuantificación de incertidumbre. Los resultados muestran que el método propuesto puede manejar efectivamente anotaciones inconsistentes mientras mantiene una alta precisión de segmentación, lo que lo hace particularmente valioso para aplicaciones clínicas.

La investigación hace contribuciones significativas al campo al proporcionar una solución robusta para manejar anotaciones inconsistentes en la segmentación de imágenes médicas, particularmente valiosa en histopatología donde las anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.

165 El enfoque propuesto muestra resultados prometedores en la reducción de costos
166 de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace
167 particularmente relevante para aplicaciones clínicas donde la segmentación precisa
168 de estructuras tisulares es crucial para el diagnóstico y la planificación del
169 tratamiento. Este trabajo abre nuevas direcciones de investigación en las áreas de
170 análisis de imágenes médicas crowdsourced, cuantificación de incertidumbre en
171 aprendizaje profundo y el desarrollo de protocolos de anotación más eficientes
172 para imágenes médicas.

173 **Palabras clave:** Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de
174 Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje
175 Profundo, Análisis de Imágenes Médicas

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- 398 **CAD** Computer-Aided Diagnosis 2, 5, 8
- 399 **CCGP** Correlated Chained Gaussian Processes 18
- 400 **CCGPMA** Correlated Chained Gaussian Processes for Multiple Annotators xxii, 18, 19,
401 53, 55–57, 59, 95
- 402 **CE** Cross Entropy 62, 65
- 403 **CGP** Chained Gaussian Processes 18, 45, 46
- 404 **CKA** Centered Kernel Alignment 44
- 405 **CNN** Convolutional Neural Networks 3, 15, 67, 95
- 406 **CR** Crowdsourcing 16, 17
- 407 **CT** Computed Tomography 1, 2, 12
- 408 **ELBO** Evidence Lower Bound 19
- 409 **EM** Expectation-Maximization 15, 20
- 410 **GCE** Generalized Cross Entropy 62, 65, 71
- 411 **GCECDL** Generalized Cross-Entropy-based Chained Deep Learning 21
- 412 **GP** Gaussian Process 46, 58
- 413 **GPMC** Gaussian Process Monte Carlo 48
- 414 **GPR** Gaussian Process Regression 48

- 415 **ISS** Image Semantic Segmentation 2, 5, 8, 10, 14, 21, 23, 37, 93, 95
- 416 **KAAR** Kernel Alignment-Based Annotator Relevance Analysis 43, 44, 46, 57
- 417 **LF** Latent Function 18
- 418 **LKAAR** Localized Kernel Alignment-Based Annotator Relevance Analysis 44–46, 57
- 419 **MAE** Mean Absolute Error 62, 65, 71
- 420 **MITs** Medical Imaging Techniques 1
- 421 **ML** Machine Learning 10, 12, 21
- 422 **MRI** Magnetic Resonance Imaging 1, 2
- 423 **MV** Majority Voting 10, 12, 20, 22
- 424 **OCR** Optical Character Recognition 12
- 425 **PET** Positron Emission Tomography 14
- 426 **RCDNN** Regularized Chained Deep Neural Network 47, 57
- 427 **ROI** Region of Interest 2, 8, 16, 31, 87
- 428 **SGPMC** Sparse Gaussian Process Monte Carlo 48
- 429 **SGPR** Sparse Gaussian Process Regression 48
- 430 **SLFM** Semi-Parametric Latent Factor Model 18, 51, 52
- 431 **SS** Semantic segmentation 3, 67
- 432 **STAPLE** Simultaneous Truth and Performance Level Estimation 13–15, 38
- 433 **TNBC** Triple-Negative Breast Cancer xxiii, xxiv, 41, 84, 87–91
- 434 **WSI** Whole Slide Imaging xxii, 1, 5, 8, 16, 29, 31, 40, 87

435

436

CHAPTER

437

ONE

438

439

INTRODUCTION

440

1.1 Motivation

441 Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced
442 significantly, with modalities like radionuclide imaging, ultrasound, **Computed**
443 **Tomography (CT)**, **Magnetic Resonance Imaging (MRI)**, and digital radiography
444 emerging over the past 50 years. Modern imaging extends beyond image
445 production to include processing, display, storage, transmission and analysis.
446 [Zhou et al., 2021]. Other **Medical Imaging Techniques (MITs)** have arose during
447 the last decades, some of them implying only the examination of certain pieces or
448 tissues instead of complete patients, like histopathological images, which are
449 images of tissue samples obtained from biopsies or surgical resections and are
450 widely used for the diagnosis of diseases like cancer through **Whole Slide Imaging**
451 **(WSI)** scanners [Rashmi et al., 2021].

452 Along with the advances in technologies for medical images acquisition,
453 computational technologies on pattern recognition and artificial intelligence have

also emerged, allowing the development of **Computer-Aided Diagnosis (CAD)** systems based on Machine Learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [Panayides et al., 2020]. One of the most common tasks in which machine learning technologies are being used in the universe of medical images is **Image Semantic Segmentation (ISS)**, which consists of assigning a label to each pixel in an image according to the object it belongs to. This is crucial for the development of **CAD** systems, as it allows the identification of **Region of Interest (ROI)** in the images, which can be used to detect and classify diseases [Azad et al., 2024].

The application of Machine Learning in medical imaging has grown significantly, including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [Brito-Pacheco et al., 2025]. Among imaging modalities, X-rays and **CT** scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. **MRI** and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [Brito-Pacheco et al., 2025]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on counted referenced terms occurrences from recent literature review, can be seen in Figure 1-1. [Yu et al., 2025], [Brito-Pacheco et al., 2025], [Ryou et al., 2025], [Hu et al., 2025], [Elhaminia et al., 2025]

Throughout the time that medical images and computational techniques have been developed, a coevolution has occurred, where the complexity of the tasks and the capabilities of the computational techniques have been increasing [Zhou et al., 2021]. Initially, these needs were covered with simple morphological

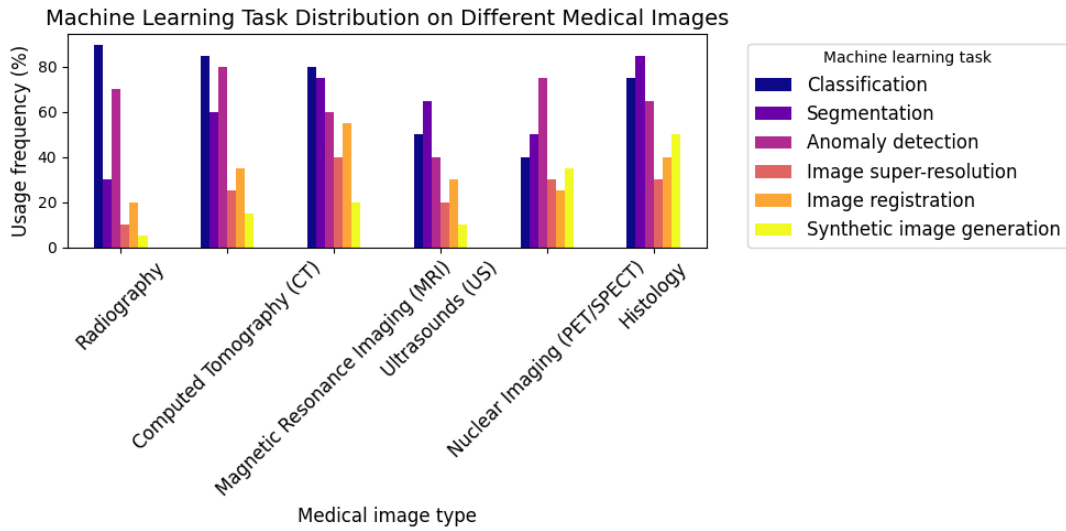


Figure 1-1 Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms.)

transformations, which implied no training process or elaborated optimization, this is an advantage in terms of development costs but has limited performance. However, as the complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the application of advanced statistical tools and Machine Learning algorithms like Support Vector Machines, Decision Trees, and SGD Neural Networks [Avanzo et al., 2024]. The coevolution of advances in medical image acquisition, computational power (i.e. Moore's law) and statistical/mathematical techniques have led to a convergence for merging state of the art algorithms with medical imaging [Shalf, 2020]. Figure 1-2 shows a timeline of coevolution between some conspicuous advances in computational pattern recognition and its medical applications in different scopes (besides medical imaging) [Avanzo et al., 2024].

Convolutional Neural Networks (CNN) have been widely used in Semantic segmentation (SS), as they have outperformed traditional Machine Learning algorithms both medical and non medical images [Xu et al., 2024] [Sarvamangala and Kulkarni, 2022]. However, most CNN architectures are deep, which introduces the requirement of large amounts of training data. This introduces a

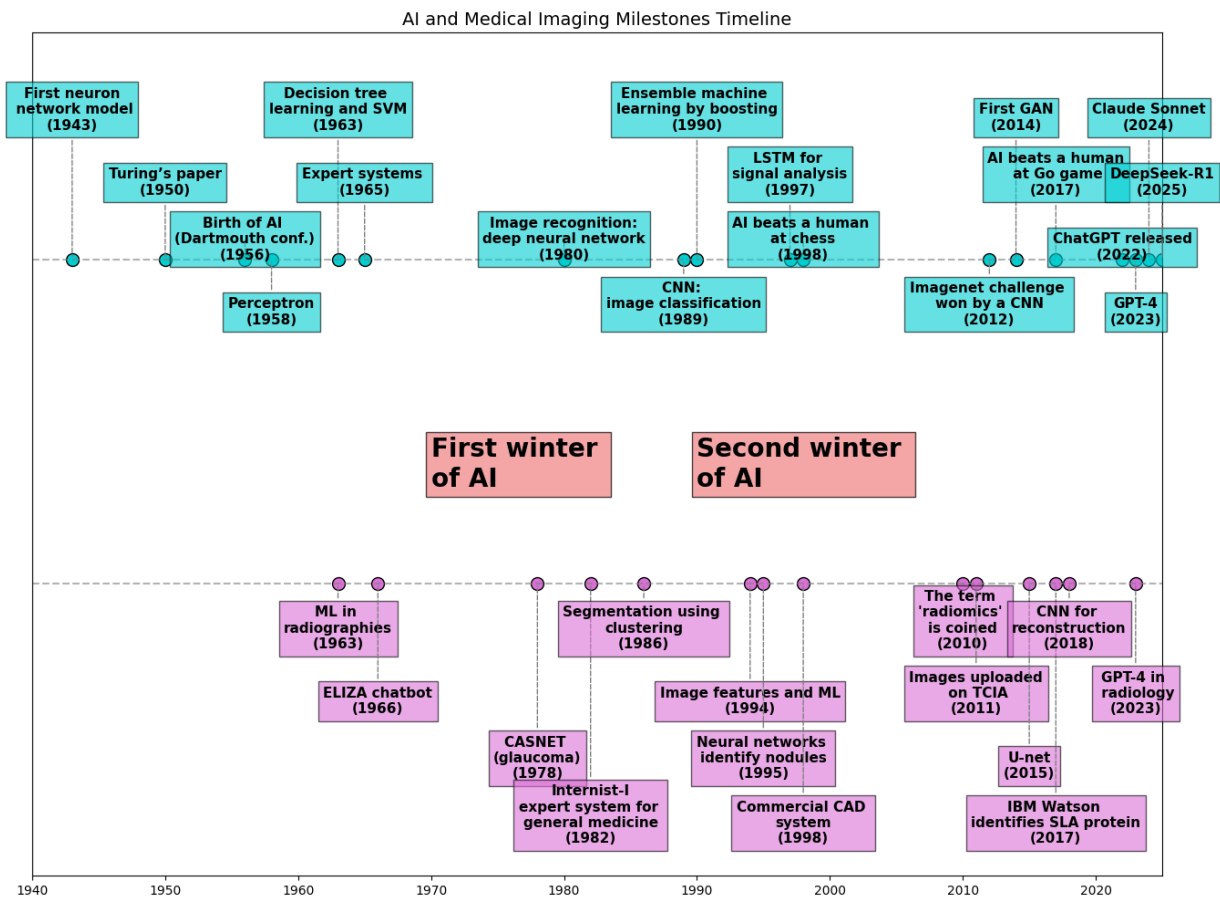


Figure 1-2 AI and Machine Learning in medical imaging recent timeline.

problem since the acquisition and annotation of medical images are expensive and time-consuming processes. This is especially true for ISS tasks, as they require pixel-level annotations, which is taxing in terms of cost, time and the logistics involved [Bhalgat et al., 2018]. Other modalities face this problem through less expensive annotation strategies like bounding boxes or anatomical landmarks for being used in a semi-supervised strategy [Shah et al., 2018].

However, many medical images datasets contain a high variability in class sizes and variations in colors, which is specially noticeable in histopathological images because of the usage of different staining techniques and other factors which can affect the color of the images (see section 2.2.3). This variability can lead to a significant loss of efficiency of Machine Learning models when using a mixed supervision strategy, as the model can be biased towards the most common classes or colors in the dataset [Shah et al., 2018].

This is where other solutions arise to tackle the problem of the weak image annotation while maintaining low costs. One of these solutions is the *crowdsourcing* strategy, which consists of having multiple annotators labeling the same image, and then combining the labels to obtain a consensus label [Lu et al., 2023]. This strategy can lead to a labeling cost reduction when different levels of expertise are combined, since the crowd may be composed of both experts and laymen, the latter being less expensive to hire [López-Pérez et al., 2023].

Recently, diagnosis, prognosis and treatment of cancer have heavily relied on histopathology, where tissue samples are obtained through biopsies or surgical resections and critical information that helps pathologists determine the presence and severity of malignancies [López-Pérez et al., 2024]. The segmentation of histopathological images enables precise identification of structures such as nuclei, glands, and tumors, which are essential for assessing disease progression and treatment response [Rashmi et al., 2021]. Accurate segmentation is particularly crucial in digital pathology, where whole-slide images (WSI) are analyzed using AI-powered CAD systems to support clinical decision-making [López-Pérez et al., 2024].

531 A major challenge in histopathological image segmentation arises from the
532 variability in annotations provided by different pathologists. Unlike natural
533 images, where object boundaries are often well-defined, histological structures
534 may have ambiguous borders, leading to inconsistencies among annotators
535 [López-Pérez et al., 2023]. Because of this, crowdsourcing labeling is one of the
536 most popular approaches, as illustrated in Figure 1-3, ¹. These discrepancies
537 highlight the need for models that can handle annotation uncertainty effectively.
538 Leveraging crowdsourcing strategies and Machine Learning techniques that infer
539 annotator reliability can enhance segmentation performance while reducing costs.

¹obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [López-Pérez et al., 2023]

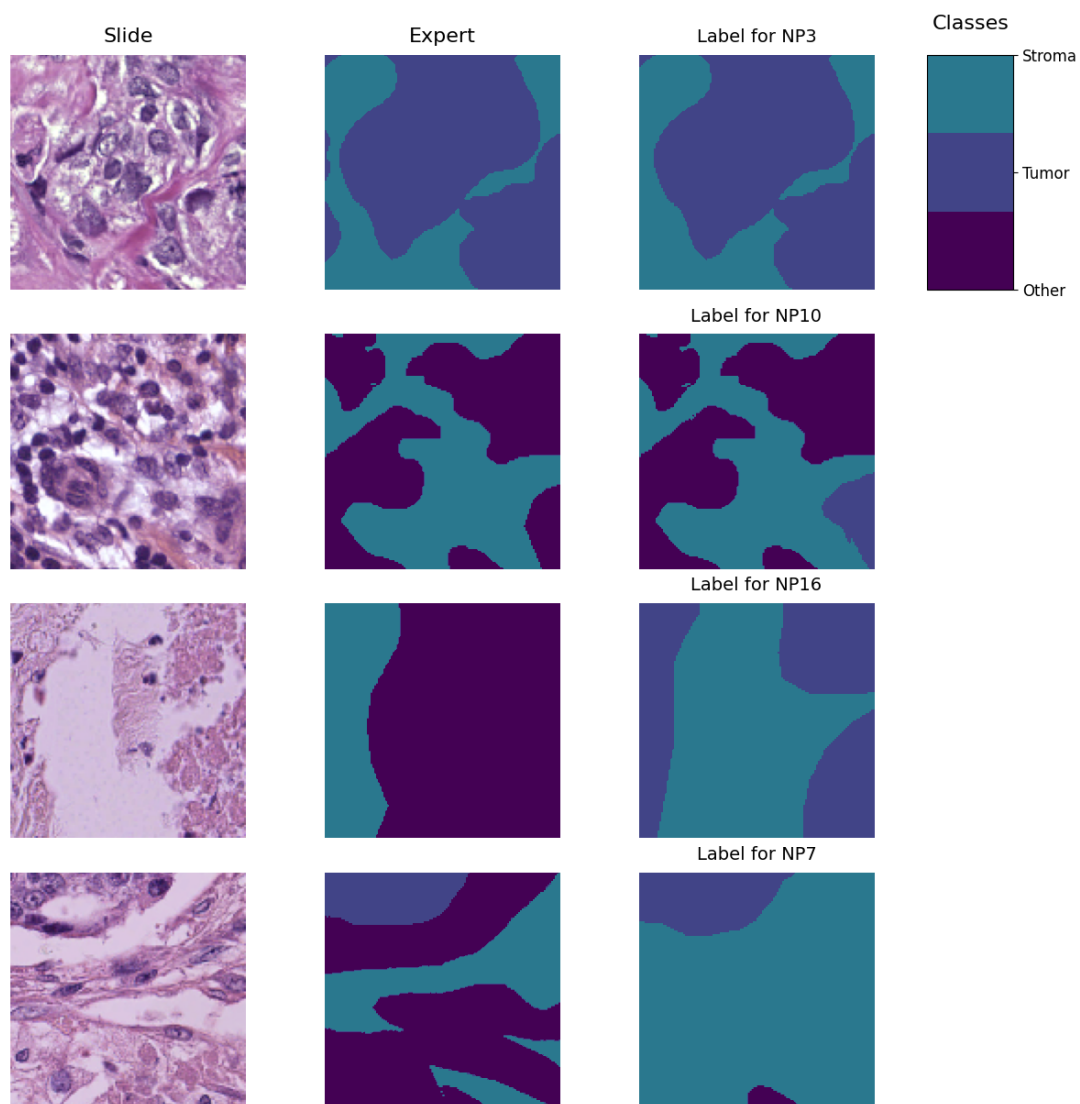


Figure 1-3 Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [Giri and Bhatia, 2024]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [Lin et al., 2024].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [Sarvamangala and Kulkarni, 2022]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments².

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering a standard format for the labeled masks [Habis, 2024]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. In the pursuit of reducing costs, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [Xu et al., 2024]. These inconsistencies are discussed in the subsections 1.2.1 and 1.2.2.

²compared to a more trivial task like image classification on ordinary an well known classes like MNIST

1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [López-Pérez et al., 2023]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy *tends* to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [Lu et al., 2023].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [López-Pérez et al., 2023]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [Banerjee et al., 2025].

Medical images frequently contain complex structures which are difficult to distinguish, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [Carmo et al., 2025]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy.

Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [Zhou et al., 2024].

When combined with long sessions, images with insufficient quality might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [Kim et al., 2024]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [Karthikeyan et al., 2023].

1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

Research Question

How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability and generalization?

1.3 State of the Art

Certainly, in general Machine Learning (ML) classification tasks³ where multiple annotators are involved, Majority Voting (MV) is by far the simplest possible approach to implement. This concept was born multiple times and divergently in

³In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

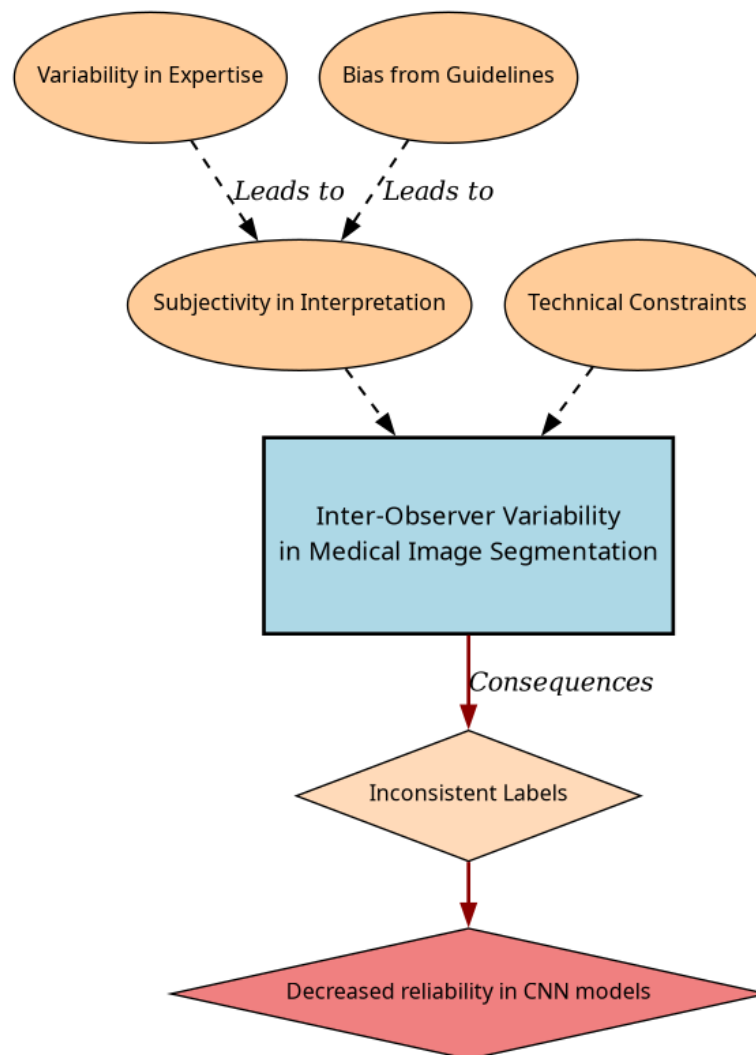


Figure 1-4 Summary diagram for problem Statement

multiple fields, but it was described as relevant for ML and pattern recognition labeling for classification in [Lam and Suen, 1997], in which the approach is exposed as simple, yet powerful. The authors describe the MV as a method that can be used to improve the accuracy of classification tasks by combining the labels of multiple annotators. The method is based on the assumption that the majority vote of the annotators is more likely to be correct than the vote of a single annotator. The authors also describe the method as a straightforward way to improve the accuracy of classification tasks without the need for complex algorithms or additional data. The authors also prove this method to deliver very similar results to more complicated approaches (Bayesian, logistic regression, fuzzy integral, and neural network) in the particular task of Optical Character Recognition (OCR). Despite its simplicity, modern solutions for delivering accurate medical image segmentation models still rely on Majority Voting at some stage. In [Elnakib et al., 2020], a majority voting strategy is used for delivering a final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and Alexnet) in CT images for Liver Tumor Segmentation. Majority voting as a technique for label aggregation is a powerful approach for its simplicity in many use cases in which the target to be labeled is not tied to an expertise related task, otherwise, the assumption of equal expertise among the labelers may introduce bias in the final label, which is not desirable in the case of highly technical annotations like medical images. In subsection 1.3.1, a review is done on literature which no longer assumes the naive approach of equal expertise among labelers and face the challenge of learning from inconsistent labels will be presented.

1.3.1 Facing Annotation Variability in Medical images

Learning from crowds approaches in general face the challenge of not having a ground truth label and hence, an intrinsic difficulty in measuring the real reliability of the labelers annotations. Some approaches assume beforehand a certain level of expertise for each labeler based on experience as an input, like in [TIAN and Zhu,

2015], which introduce the concept of max margin majority voting, using the reliability vector as weights for the binary and multiclass classifier. The crowdsourcing margin is the minimal difference between the aggregated score of the potential true label and the scores for other alternative labels. Accordingly, the annotators' reliability is estimated as generating the largest margin between the potential true labels and other alternatives. The problem introduced in this approach is assuming an stationary reliability per expert across the whole input space, which is imprecise since annotators performance may change between different tasks or even between different regions of the same image.

STAPLE Mechanism

The Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm, introduced in [Warfield et al., 2004] is a probabilistic framework that estimates a hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

The STAPLE algorithm's goal is to maximize the log likelihood function:

$$(\hat{\mathbf{p}}, \hat{\mathbf{q}}) = \arg \max_{\mathbf{p}, \mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \quad (1-1)$$

Where $\mathbf{D}$ is the set of segmentations provided by the raters, $\mathbf{T}$ is the hidden true segmentation, p is the sensitivity and q is the specificity of the raters.

This is achieved by using the Expectation-Maximization algorithm to maximize the log likelihood function, which is done iteratively with step computations:

$$\begin{aligned}
(p_j^{(k)}, q_j^{(k)}) = \arg \max_{p_j, q_j} & \sum_{i: D_{ij}=1} W_i^{(k-1)} \ln p_j \\
& + \sum_{i: D_{ij}=1} (1 - W_i^{(k-1)}) \ln(1 - q_j) \\
& + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\
& + \sum_{i: D_{ij}=0} (1 - W_i^{(k-1)}) \ln q_j.
\end{aligned} \tag{1-2}$$

656 The weight terms $W_i^{(k-1)}$ and $(1 - W_i^{(k-1)})$ represent the current estimates of the
657 probability that voxel i belongs or does not belong to the true segmentation from
658 the previous E-step. The log-likelihood contributions $\ln p_j$, $\ln(1 - q_j)$, $\ln(1 - p_j)$, and
659 $\ln q_j$ account for true positives, false positives, false negatives, and true negatives,
660 respectively, weighted by the current estimate of true segmentation probabilities.

661 The capacity of STAPLE to accurately estimate the true segmentation, even in the
662 presence of a majority of raters generating correlated errors, was demonstrated,
663 which makes it theoretically a strong choice for setting a ground-truth in binary or
664 multiclass medical ISS tasks.

665 The popularity and performance of STAPLE has led to its usage in modern
666 applications medical image, 3d spatial images due to its assumption of decision
667 space being based on voxel-wise decisions, like the authors in [Grefve et al., 2024]
668 which applied the algorithm on Positron Emission Tomography (PET) images.
669 Other authors still rely heavily on STAPLE for setting a ground truth consensus for
670 histopathological images, like [Qiu et al., 2022].

671 However, the STAPLE algorithm has some limitations. It assumes independent
672 rater errors, which may not hold in practice, leading to biased estimates. STAPLE
673 is also sensitive to low-quality annotations, potentially degrading final
674 segmentations if the weights are not initialized correctly. The algorithm tends to
675 over-smooth results, blurring fine details, and struggles with multi-class

segmentation. Computationally, it is expensive due to its iterative Expectation-Maximization (EM) approach. Additionally, STAPLE cannot correct systematic biases in annotations and depends on initial estimates, impacting accuracy. Lastly, the estimated performance levels lack interpretability, making it difficult to assess annotator reliability effectively.

Finally, while STAPLE proves valuable for aggregating existing labels and supporting other methodologies, its primary limitation lies in its inability to generate annotations for novel, unlabeled images. This restricts its applicability to scenarios where multiple annotations already exist, rather than serving as a standalone solution for new image segmentation tasks.

U-shaped CNNs

Since the introduction of U-Net [Ronneberger et al., 2015] in 2015 for biomedical image segmentation, U-shaped CNNs have become a prevalent architecture in medical image segmentation tasks. The U-Net's success stems from its ability to capture both global and local information through its contracting and expanding paths, making it particularly effective for complex and heterogeneous structures, even with limited annotated data. This architecture has been successfully applied to various medical image segmentation tasks, including organ segmentation, tumor segmentation, and brain structure segmentation [Azad et al., 2024].

The U-Net architecture consists of a symmetric encoder-decoder structure with skip connections. The encoder path progressively reduces spatial dimensions while increasing feature channels through a series of convolutional and max-pooling layers, capturing high-level semantic information. The decoder path uses transposed convolutions to gradually recover spatial resolution while reducing feature channels. Skip connections between corresponding encoder and decoder layers preserve fine-grained details by concatenating high-resolution features from the encoder with upsampled features in the decoder, enabling precise localization of structures.

In [López-Pérez et al., 2024] two networks are trained for delivering a final segmentation. One network is trained to estimate the annotators reliability and another one is trained to segment the image. The first network is a Deep Neural Network that takes as input features of image and the one-hot encoded identifiers of the labelers and outputs a reliability map across the image feature space. This map is then used to weight the contribution of each annotator to the final segmentation. The second network is the U-Net used for segmentation.

In this approach, it is assumed that the images are labeled for at least one labeler and not all of them, which is closer to a real world scenario, in which images with variability in the amount of annotations per patch is common. Hence, the input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \dots, N; r \in R_n\}, \quad (1-3)$$

Where every $\mathbf{x}_n$ is an input patch from a ROI in one WSI, $\tilde{\mathbf{y}}_n$ is the noisy annotation from the r labeler, N is the number of patches in the dataset and $R_n \subset \{1, \dots, R\}$ is the set of labelers that annotated the image $\mathbf{x}_n$.

The authors then assume the annotator network to deliver a reliability map $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$ with different dimensions:

- **Crowdsourcing (CR) global:** a single reliability vector per labeler with dimensions C which represent global reliability of the labeler across all input space.
- **CR image:** a single reliability vector per image per labeler with dimensions C which represent local reliability of the labeler across the image.
- **CR pixel:** a reliability matrix per image per labeler, with dimensions C which represent local reliability of the labeler across all the pixels in the image.

727 These differences in dimensions are determined by the feature extraction space
 728 from segmentation network which feed the input of the annotator network, which
 729 the authors vary for experimentation purposes.

730 Let $\mathbf{p}_\theta(\mathbf{x}_n)$ be the estimated latent (ground truth) segmentation produced by the
 731 segmentation U-Net network. The estimated segmentation probability mask for
 732 each annotator is then computed as the product:

$$\mathbf{p}_{\theta,\phi}^{(r)}(\mathbf{x}_n) := \mathbf{A}_\phi^{(r)}(\mathbf{x}) \odot \mathbf{p}(\mathbf{x}_n), \quad (1-4)$$

733 where $\odot$ is the element-wise product and ϕ and θ are the parameters of the
 734 annotator network and the segmentation UNet network, respectively, being the
 735 latter initialized with a ResNet34 backbone pre-trained on ImageNet.

736 The authors propose a loss function involving cross-entropy and a trace based
 737 regularization on the reliability map, originally proposed in [Zhang et al., 2020]
 738 which combined, looks like:

$$\mathcal{L}(\theta, \phi) := \sum_{n=1}^N \sum_{r=1}^R \mathbb{I}(\tilde{\mathbf{y}}_n^{(r)} \in R_n) \cdot \left[\text{CE} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \cdot \mathbf{p}_\theta(\mathbf{x}_n), \tilde{\mathbf{y}}_n^{(r)} \right) + \lambda \cdot \text{tr} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \right) \right] \quad (1-5)$$

739 Being $\mathbb{I}$ the indicator function, CE the cross-entropy loss, and λ the regularization
 740 parameter.

741 When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves
 742 a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching
 743 expert-supervised performance (0.7723). The CR image reliability modeling
 744 proved most effective, on the other hand CR pixel, while potentially offering
 745 finer-grained reliability estimation, requires significantly more training data.

746 Despite the proven performance of the approach by the authors, solving the problem
 747 of multiple labelers with two networks can be overwhelming for the optimization
 748 process, requiring large amounts of annotated data to properly codify the annotators
 749 spatial reliabilities, which could be managed by a single model with an appropriate
 750 loss function.

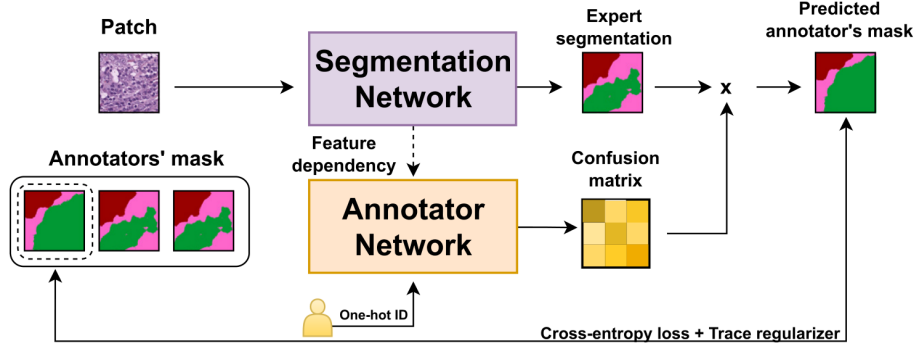


Figure 1-5 Proposed framework for the approach in [López-Pérez et al., 2024].

Bayesian Models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [Gil-González and Álvarez Meza, 2023] the authors propose a novel approach based on Gaussian Processes for modeling the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA), a framework based on the well known Chained Gaussian Processes (CGP). CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the Correlated Chained Gaussian Processes (CCGP) thus, the authors introduce the CCGP to model correlations between the GP latent functions, which are supposed to be generated from a Semi-Parametric Latent Factor Model (SLFM):

$$f_j(\mathbf{x}_n) = \sum_{q=1}^Q w_{j,q} \mu_q(\mathbf{x}_n), \quad (1-6)$$

where $f_j : \mathcal{X} \rightarrow \mathbb{R}$ is a Latent Function (LF), $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$ with $k_q : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ being a kernel function, and $w_{j,q} \in \mathbb{R}$ is a combination coefficient ($Q \in \mathbb{N}$). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u | \mathbf{X}) = p(\mathbf{y} | \boldsymbol{\theta}) \prod_{j=1}^J p(\mathbf{f}_j | \mathbf{u}) p(\mathbf{u}), \quad (1-7)$$

766 where $\mathbf{y}$ is the vector of noisy labels, $\hat{\mathbf{f}}$ is the vector of latent functions, u represents
767 the inducing points, and $\mathbf{X}$ is the input data.

768 Combined with inducing-variables based methods for sparse GP approximations,
769 and maximizing an **Evidence Lower Bound (ELBO)** for the estimation of the
770 variational parameters, the authors reach a model whose variational expectations
771 are not analytically tractable, and hence, the authors derive a Gaussian-Hermite
772 quadrature approach.

773 Finally, the authors extend this approach for being applied to classification and
774 regression, reaching the only known approach to involve chained gaussian
775 processes in multiple annotators classification and regression tasks while
776 preserving the interdependencies among the annotators, and also outperforming
777 GPC-MV⁴, MA-LFC-C⁵, MA-DGRL⁶, MA-GPC⁷, MA-GPCV⁸, MA-DL⁹, KAAR¹⁰.

778 **CCGPMA** on itself proposes a good approach for handling label noise and
779 uncertainty in the labelers for regression and classification tasks, while also
780 preserving the interdependencies among the annotators, however, it does not face
781 the image segmentation problem, which is the main focus of this work. Besides,
782 handling so many latent functions during the optimization process is
783 computationally expensive, making it on itself infeasible for large and high
784 resolution datasets.

⁴A GPC using the MV of the labels as the ground truth.

⁵A LRC with constant parameters across the input space.

⁶A multi-labeler approach that considers as latent variables the annotator performance.

⁷A multi-labeler GPC, which is an extension of MA-LFC.

⁸An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.

⁹A Crowd Layer for DL, where the annotators' parameters are constant across the input space.

¹⁰A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

1.3.2 Facing Noisy Annotations and Low-Quality Data

The problem of low-quality data and noisy annotations has been tackled with various strategies. One such approach is the use of deep learning models that incorporate loss functions designed to mitigate the effects of unreliable labels. Traditional methods such as *MV* or *EM* have been widely used for aggregating multiple annotators' inputs. However, they assume a homogeneous reliability of annotators, which may not hold in real-world scenarios.

Loss Functions in Deep Learning Models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [Zhang and Sabuncu, 2018]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [Zhao et al., 2020]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

Generalized Cross-Entropy for Multiple Annotators Classification

A more recent approach was proposed by [Triana-Martinez et al., 2023], introducing a Generalized Cross-Entropy-based Chained Deep Learning (GCECDL) framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator’s reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, GCECDL accounts for the dependencies among annotators while encoding their non-stationary behavior across different data samples. Their experiments on multiple datasets demonstrated superior predictive performance compared to state-of-the-art methods, particularly in cases where annotations were highly inconsistent.

The strategy of the authors effectively unlocks the potential of ML models to handle low-quality data and noisy annotations, but it is bounded to classifications tasks only, not being by itself applicable to segmentation tasks. The TGCE equation for handling multiple annotators is defined as:

$$\text{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x) \frac{1 - (\tilde{C})^q}{q}, \quad (1-8)$$

where $\tilde{\lambda}_x$ represents the annotator reliability, $\tilde{C}$ is a constant, q is a parameter that controls the balance between MAE and CE behavior, $\mathbf{y}$ is the annotation vector, and $f(\mathbf{x})$ is the model prediction. This approach is more deeply discussed in chapter 4.

1.4 Aims

1.4.1 General Aim

The main purpose of this work is to adaptively infer the best possible segmentation in ISS tasks for medical images through a novel approach, without needing separate

Table 1-1 Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

Approach	Key Characteristics	Advantages	Limitations
MV [Lam and Suen, 1997]	Simple aggregation of multiple annotators' labels	- Simple to implement - No complex algorithms required - Proven effective in many cases	- Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks
STAPLE [Warfield et al., 2004]	Probabilistic framework estimating true segmentation and rater reliability	- Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors	- Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability
U-Net based approaches [Ronneberger et al., 2015]	Deep Learning architecture with encoder-decoder structure	- Captures both global and local information - Effective for complex structures - Works well with limited data	- Requires large amounts of training data - Complex optimization process - May need specialized loss functions
Bayesian Models (CCGPMA) [Gil-Gonzalez et al., 2023]	Gaussian Processes modeling annotator reliability	- Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression	- Computationally expensive - Not directly applicable to segmentation - Complex optimization process
Generalized Cross-Entropy (GCE) [Zhang and Sabuncu, 2018]	Loss function balancing MAE and CE	- Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies	- Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation

834 inputs about the labeler while offering an interpretable reliability space per labeler
835 and properly facing the performance inconsistency across the annotators space and
836 the variability of images quality.

837 1.4.2 Specific Aims

- 838 • To develop a novel loss function for ISS tasks in medical images, which
839 accurately infers the underlying ground truth segmentation mask without
840 needing separate inputs about the labeler performance.
- 841 • To introduce a tensor map which codifies the reliability of each labeler based
842 only on the inputs, allowing the model to implicitly weigh the labelers based
843 on their performance across the mask and classes space.
- 844 • To develop and test a deep learning model for ISS tasks in medical images,
845 which can learn from inconsistent labels and improve the segmentation
846 performance compared to other solutions in state of the art.

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848

CHAPTER

849

TWO

850

851

CONCEPTUAL PRELIMINARIES

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2.1 Modern Concept of Digital Image

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A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function $f(x, y)$ that maps spatial coordinates (x, y) to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

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2.1.1 Types of Digital Images

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Grayscale Images

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864

Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix I of size $M \times N$, where each element $I(i, j)$ represents the intensity at position (i, j) . The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

865 **Color images**

866 Color images extend the grayscale concept by representing each pixel with multiple
867 channels, typically Red, Green, and Blue (RGB). A color image can be represented
868 as a 3D matrix I of size $M \times N \times 3$, where $I(i, j, k)$ represents the intensity of the
869 k -th color channel at position (i, j) . Other color spaces like HSV (Hue, Saturation,
870 Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different
871 applications.

872 **Multispectral images**

873 Multispectral images capture information across multiple wavelength bands
874 beyond the visible spectrum. These images can be represented as a 3D matrix I of
875 size $M \times N \times B$, where B is the number of spectral bands. Each band $I(i, j, b)$
876 represents the intensity at position (i, j) for the b -th spectral band. This
877 representation is particularly useful in medical imaging, remote sensing, and
878 several scientific applications.

879 **3D Images and Volumetric Data**

880 Three-dimensional images extend the concept of pixels to voxels (volume elements).
881 A 3D image can be represented as a 3D matrix V of size $M \times N \times D$, where D
882 represents the depth dimension. Each voxel $V(i, j, k)$ represents the intensity at
883 position (i, j, k) in the 3D space. This representation is fundamental in medical
884 imaging (CT, MRI), scientific visualization, and computer graphics.

885 **2.1.2 Mathematical Representations**

886 The mathematical foundation of digital images relies on several key concepts:

- 887 • **Sampling:** The process of converting a continuous image into a discrete
888 representation. According to the Nyquist-Shannon sampling theorem, the
889 sampling frequency must be at least twice the highest frequency present in
890 the image to avoid aliasing.
- 891 • **Quantization:** The process of converting continuous intensity values into
892 discrete levels. The number of quantization levels determines the image's
893 bit depth and affects its quality and storage requirements.
- 894 • **Resolution:** The number of pixels per unit length in an image, typically
895 measured in pixels per inch (PPI) or dots per inch (DPI).
- 896 • **Dynamic range:** The ratio between the maximum and minimum measurable
897 light intensities in an image, often expressed in decibels (dB).

898 The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \quad (2-1)$$

899 where $I(x, y)$ is the digital image, $f(i, j)$ represents the intensity values, and $\delta(x -$
900 $i, y - j)$ is the Kronecker delta function [Duarte and Baraniuk, 2012].

901 For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \quad (2-2)$$

902 where I_R , I_G , and I_B represent the red, green, and blue channels respectively.

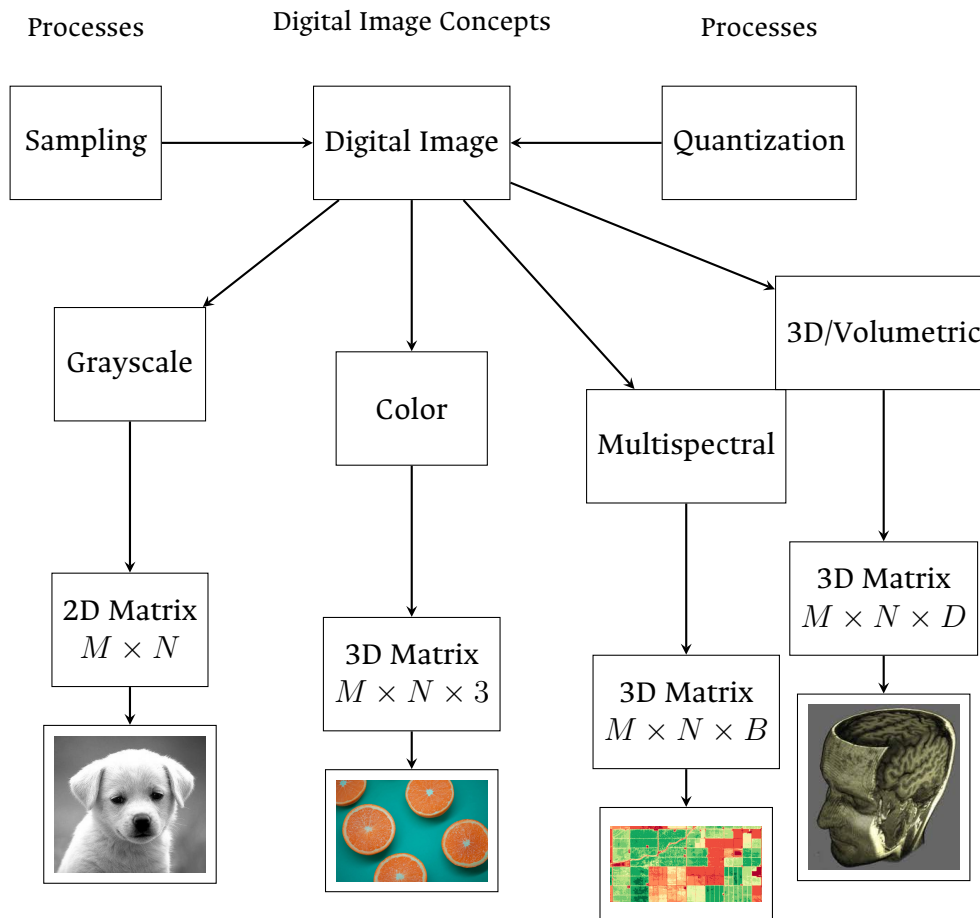


Figure 2-1 Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [Amgad et al., 2019]. This process has evolved significantly over the past few decades, as shown in Figure 2-2.

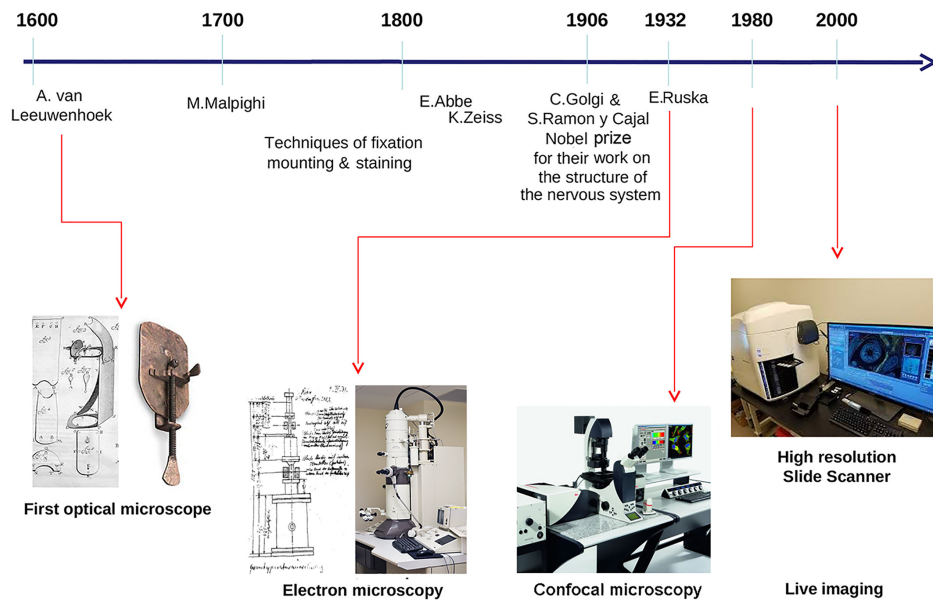


Figure 2-2 Histology evolution timeline. (Image from [Mazzarini et al., 2021]).

2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern WSI scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched

915 together to create a seamless digital image [Hu et al., 2025]. These systems
916 incorporate high-resolution objectives with magnifications ranging from 20x to
917 40x, precise motorized stages for accurate slide positioning, automated focus
918 systems to maintain image quality, and high-quality cameras equipped with large
919 sensor arrays. The resulting digital slides can reach sizes of several gigabytes,
920 containing billions of pixels that capture the microscopic details of tissue samples
921 [Hu et al., 2025]. Figure 2-3 shows a whole slide imaging system by Omnyx for
922 slide digitization and a comprehensive digital pathology interface from Omnyx
923 designed to streamline pathologists' diagnostic workflow [Farahani et al., 2015].



Figure 2-3 (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology scan visualization from Omnyx designed to streamline pathologists' diagnostic workflow. (From [Farahani et al., 2015]).

2.2.2 Regions of Interest (ROI)

In digital histopathology, **Region of Interests (ROIs)** are specific areas within a whole slide image that contain diagnostically relevant information. The importance of **ROIs** lies in their ability to focus computational analysis on relevant areas, reduce computational complexity in automated systems, facilitate targeted diagnosis and research, and enable efficient storage and transmission of critical information.

Selecting **ROIs** is a fundamental step in medical image analysis across all imaging modalities, where an ROI represents a subset of an image appropriate for the intended analysis and is typically identified manually by experts [Hossain et al., 2023]. In modern pathology, the analysis involves processing multidimensional and high-resolution **WSI** tiles automatically, containing an overwhelming quantity of structural and functional information. Despite recent improvements in computing capacity, analyzing such extensive data remains challenging but vital for accurate analysis. Automatic ROI detection can significantly reduce the number of pixels to be processed, speed up the analysis, improve accuracy, and reduce dependency on pathologists [Hossain et al., 2023].

The challenge of acquiring image domain annotations for ROI detection is particularly significant due to the knowledge and commitment required of expert pathologists. Pathologists often identify regions in whole slide images with diagnostic relevance rather than examining the entire slide, with a positive correlation between the time spent on these critical image regions and diagnostic accuracy [Ghezloo et al., 2025]. This selective viewing behavior highlights the importance of developing automated systems that can replicate pathologists' diagnostic patterns.

Recent advances in deep learning techniques have offered improvements in computer-aided diagnosis systems for ROI detection. Innovative approaches include generating heatmaps to represent pathologists' viewing patterns during

diagnosis and using these to guide deep learning architectures during training [Ghezloo et al., 2025]. Such systems can integrate pathologists' domain expertise to enhance region of interest detection without needing individual case annotations, potentially surpassing traditional approaches based on color and texture image characteristics.

The clinical application of automated ROI detection has been demonstrated in specific diagnostic contexts, such as Human Epidermal Growth Factor Receptor 2 (HER2) grading for breast cancer patients. Existing HER2 grading relies on manual ROI selection, which is tedious, time-consuming, and suffers from inter-observer and intra-observer variability [Hossain et al., 2023]. Studies have found that HER2 grade changes with ROI selection, emphasizing the critical importance of accurate and consistent ROI detection methods. Advanced approaches using Vision Transformer architectures have shown promising results, achieving high accuracy rates for ROI detection while significantly reducing processing time and improving diagnostic agreement with clinical scores [Hossain et al., 2023].

These automated ROI detection methods not only enhance diagnostic accuracy and efficiency but also streamline annotation processes and aid in the training of new pathologists. By incorporating pathologists' expertise through heatmap generation and advanced deep learning architectures, these systems demonstrate the potential to revolutionize digital pathology workflows while maintaining the quality and reliability of diagnostic assessments [Ghezloo et al., 2025] (Figure 2-4).

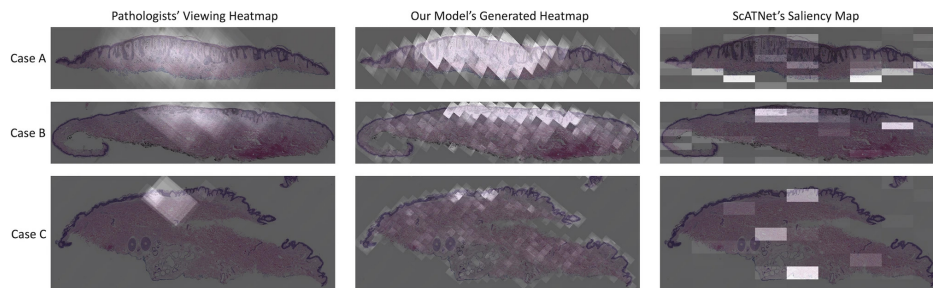


Figure 2-4 ROI heatmaps measured experimentally from pathologists view patterns and estimated heatmap from a U-Net model from [Ghezloo et al., 2025].

973 **2.2.3 Staining Techniques**

974 Histopathological analysis relies heavily on various staining techniques to enhance
975 the visibility of different tissue components and cellular structures. The choice of
976 staining method depends on the specific diagnostic requirements and the type of
977 tissue being examined.

978 **Hematoxylin and Eosin (H&E)**

979 Hematoxylin and Eosin (H&E) staining is the most widely used technique in
980 histopathology, particularly in breast cancer diagnosis [Pan et al., 2021]. This
981 staining method provides essential visualization through two components:
982 hematoxylin, which stains cell nuclei blue/purple to highlight nuclear morphology,
983 and eosin, which stains cytoplasm and extracellular matrix pink/red to reveal
984 tissue architecture.

985 The popularity of H&E staining in breast cancer histopathology stems from its
986 ability to clearly visualize tumor architecture and growth patterns, distinguish
987 between different types of breast cancer, identify important diagnostic features
988 like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast
989 cancer, H&E staining finds extensive application across various medical specialties
990 including general pathology, dermatology, gastroenterology, neurology, and
991 oncology.

992 **Special Stains**

993 In addition to H&E, various special stains are used for specific diagnostic purposes.
994 Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a
995 crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor
996 (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR)
997 staining for assessing progesterone receptor status, Human Epidermal Growth

Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and Ki67 staining for measuring cellular proliferation rates. These markers are particularly crucial in breast cancer diagnosis and treatment planning, as they help determine the molecular subtype of the cancer and guide personalized therapeutic approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for highlighting carbohydrates and basement membranes, Masson's Trichrome for distinguishing between collagen and muscle fibers, and silver stains for detecting microorganisms and nerve fibers. These specialized staining techniques complement H&E by providing additional diagnostic information that is crucial for accurate diagnosis and treatment planning [Weitz et al., 2023]. Examples of these staining techniques are shown in Figure 2-8.

2.3 Deep Learning Fundamentals

Deep Learning has emerged as a powerful subset of Machine Learning, revolutionizing the field of artificial intelligence. Its roots can be traced back to the early development of artificial neural networks in the 1940s and 1950s, with significant milestones including the perceptron in 1958 and the backpropagation algorithm in the 1980s. However, it wasn't until the early 21st century, with the advent of more powerful computational resources and the availability of large datasets, that deep learning truly began to flourish [Werbos, 1990].

2.3.1 Learning Paradigms

Machine Learning systems can be categorized into three main learning paradigms. The most common approach is supervised learning, where models learn from labeled data by mapping inputs to known outputs. This paradigm requires a large amount of labeled training data, which can be expensive and time-consuming to

acquire. Semi-supervised learning offers a hybrid approach that leverages both labeled and unlabeled data [Alloghani et al., 2020], proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance [Alloghani et al., 2020], making it valuable for tasks like clustering and dimensionality reduction. Being deep learning a subset of Machine Learning, it is based on the same paradigms. An illustration of the three main learning paradigms is shown in Figure 2-5.

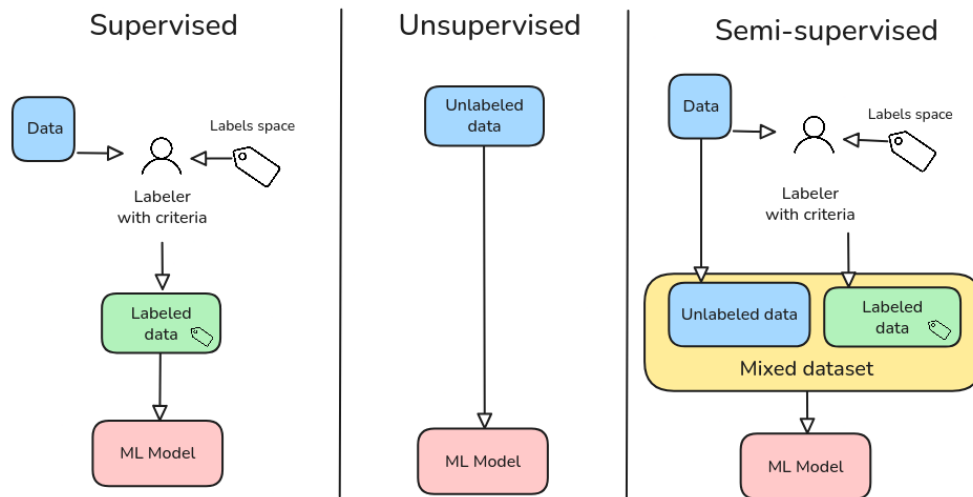


Figure 2-5 Common learning paradigms.

2.3.2 Architecture and Training

Deep Learning architectures are characterized by their layered structure, where each layer progressively extracts and transforms features from the input data [Taye, 2023]. The early layers typically focus on low-level feature extraction, such as edges, textures, and basic patterns in the case of image processing [Taye, 2023]. As information flows through the network, middle layers combine these basic features into more complex representations. The final layers perform high-level reasoning and make the ultimate predictions or classifications.

1038 The training process commonly employs gradient descent as its primary
1039 optimization strategy, which iteratively adjusts the model's parameters to
1040 minimize a loss function. However, other optimization algorithms like Adam,
1041 RMSprop, and evolutionary strategies [Soydaner, 2020] [Bingham et al., 2020], can
1042 also be effective for training deep networks. The loss function serves as a crucial
1043 component of the learning process in most optimization techniques, quantifying
1044 how well the model's predictions match the actual targets. By providing a measure
1045 of the model's performance, the loss function guides the chosen optimization
1046 process, enabling the network to learn meaningful patterns from the training data.

1047 2.3.3 Challenges and Solutions

1048 Despite their power, deep learning systems face several significant challenges. One
1049 of the most prominent issues is overfitting, where models may memorize training
1050 data instead of learning generalizable patterns [Roelofs et al., 2019]. This challenge
1051 is typically addressed through various regularization techniques such as dropout,
1052 L1/L2 regularization, and early stopping [Roelofs et al., 2019]. Another critical
1053 challenge is the substantial data requirements; deep learning models often need
1054 massive amounts of training data to achieve good performance [Roelofs et al.,
1055 2019], which can be a limiting factor in many applications. Additionally, the
1056 complex, layered nature of deep learning models makes them difficult to interpret,
1057 often referred to as "black boxes". This lack of transparency can be particularly
1058 problematic in critical applications where understanding the decision-making
1059 process is essential.

1060 2.3.4 Deep Learning Frameworks

1061 The development of powerful open-source frameworks has significantly
1062 accelerated deep learning research and applications. TensorFlow, developed by

Google, provides a comprehensive ecosystem for building and deploying Machine Learning models [Abadi et al., 2016]. PyTorch, created by Facebook’s AI Research lab, offers dynamic computation graphs and has become particularly popular in research settings [Paszke et al., 2019]. Caffe, known for its speed and modularity, is widely used in computer vision applications [Jia et al., 2014].

These frameworks have democratized deep learning by providing efficient implementations of common operations, automatic differentiation for gradient computation, and GPU acceleration for faster training. They also offer pre-trained models and transfer learning capabilities, along with active communities for support and knowledge sharing. The combination of these frameworks with modern hardware has enabled researchers and practitioners to develop increasingly sophisticated models, pushing the boundaries of what’s possible in artificial intelligence. TensorFlow and PyTorch have emerged as the two most prominent frameworks in the deep learning landscape, as shown in Figure 2-6, which presents data from Google Trends over the last five years (as of April 2025).

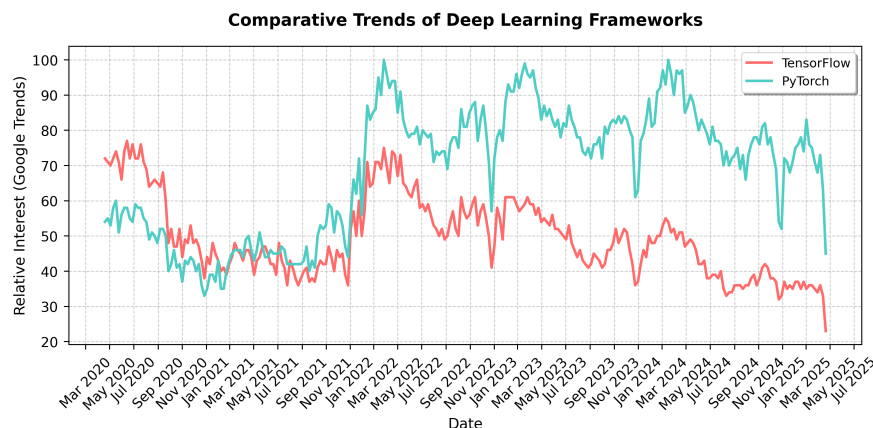


Figure 2-6 Comparative Trends of the top two most popular Deep Learning Frameworks.

2.4 Datasets and Data Sources

Throughout the development of this work, multiple datasets were used for evaluation of ISS models. The common elements of all these datasets are that they

contain RGB images and are crowdsourced with multiple labelers, where not necessarily all labeler label all images.

As it has been mentioned in Chapter 1, the main goal of this work is mainly focused on crowdsourced histopathology images semantic segmentation, however, these datasets present the following challenges:

- Distribution of segmentation labels is not uniform across the image, since some tissues and structures are more common than others.
- Visualization of performance of the models in debug time (like per epoch analysis) is not simple for non experts in the subject, which makes it hard to evaluate whether the model is overfitting or not at a glance.

For these reasons, multiple datasets were created this work in the pursuit of an initial evaluation of performance of the models against more traditional and familiar images before the focus on histopathology images. Once a satisfactory performance was achieved, the focus was shifted to histopathology images and further tunings on the models were performed if needed.

In any case, both the emulated noisy annotations datasets and the histopathology datasets somehow contained ground truth aggregation, either from the original source (in the case of emulated noisy annotations), the expert annotation (if available) or from the aggregation of multiple labelers ¹.

2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section 5.5.1, the Oxford-IIIT Pet Dataset [Parkhi et al., 2012] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the ‘tensorflow_datasets’ package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: ‘background’, ‘border’, and ‘pet’. Figure 2-7 shows an example of the annotations in the original dataset.

¹STAPLE in the case of histopathology datasets with no expert annotations available

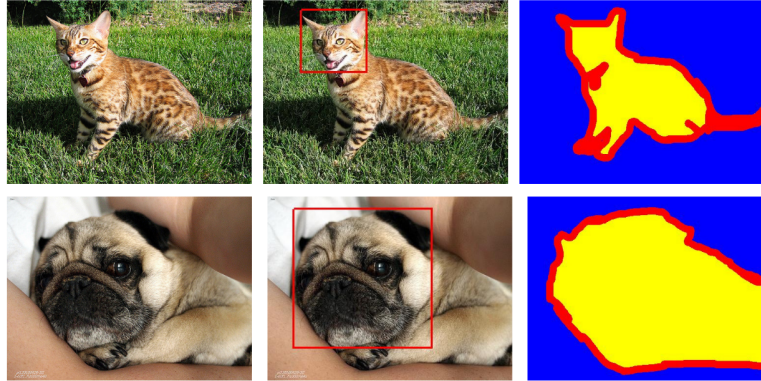


Figure 2-7 Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

2.4.2 Real histopathology datasets

Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [Weitz et al., 2023] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER (Estrogen Receptor), PGR (Progesterone Receptor), HER2 (Human Epidermal Growth Factor Receptor 2), and KI67 (Ki67 Proliferation Marker), which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section 2.2.3).

The dataset's relevance to our work stems from several key aspects. The matched H&E and IHC stains allows for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a

1125 real-world clinical setting. The inclusion of routine diagnostic cases makes the
 1126 dataset representative of actual clinical practice, where variations in staining
 1127 quality and tissue preparation are common. Furthermore, the multiple biomarker
 1128 stains (ER, PGR, HER2, KI67) enable to study how different tissue characteristics
 1129 affect segmentation performance and annotator agreement.

1130 This dataset serves as an ideal testbed for our crowdsourced segmentation
 1131 approach. It allows us to evaluate how different staining modalities affect
 1132 annotator performance and agreement, while also providing insights into the
 1133 relationship between tissue characteristics and segmentation difficulty. The
 1134 dataset's comprehensive nature enables validation of our models' performance
 1135 across different biomarker expressions and assessment of the generalizability of
 1136 our approach to real-world clinical data.

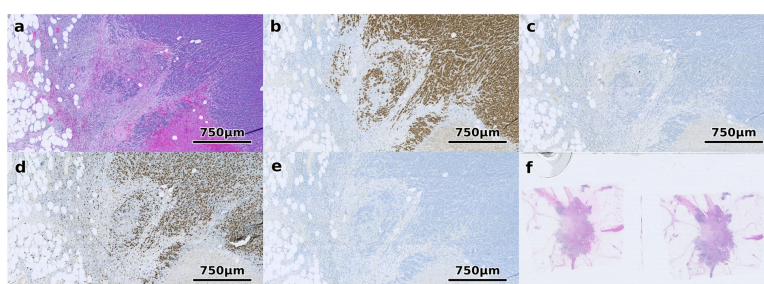


Figure 2-8 Different staining techniques obtained from multi-stain breast cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

1137 Structured Crowdsourcing Dataset for Histology Images

1138 The dataset presented by [Amgad et al., 2019] is particularly relevant to our work
 1139 as it represents one of the first systematic studies of crowdsourced annotations in
 1140 histopathology. The authors recruited 25 participants with varying levels of
 1141 expertise (from senior pathologists to medical students) to delineate tissue regions
 1142 in 151 breast cancer slides using the Digital Slide Archive platform, resulting in
 1143 over 20,000 annotated tissue regions. The annotation efforts are focused on

1144 Triple-Negative Breast Cancer (TNBC), an aggressive genomic subtype that
1145 comprises 15% of breast cancer cases [Plasilova et al., 2016].

1146 Key aspects of this dataset make it valuable for our work. The systematic
1147 evaluation of inter-participant discordance revealed varying levels of agreement
1148 across different tissue classes, with low discordance for tumor and stroma, and
1149 higher discordance for more subjectively defined or rare tissue classes. The
1150 inclusion of feedback from senior participants helped in curating high-quality
1151 annotations, demonstrating that fully convolutional networks trained on these
1152 crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The
1153 dataset also provides evidence that the scale of annotation data significantly
1154 improves image classification accuracy.

1155 This dataset is particularly valuable for our work because it provides crucial
1156 insights into how annotator expertise affects segmentation quality. It
1157 demonstrates the feasibility of using crowdsourced annotations for training
1158 accurate segmentation models, showing that even with varying levels of expertise,
1159 aggregated annotations can produce reliable ground truth. The dataset includes a
1160 systematic analysis of inter-annotator agreement, which is crucial for
1161 understanding the challenges in crowdsourced histopathology segmentation and
1162 informing the development of more robust segmentation approaches.

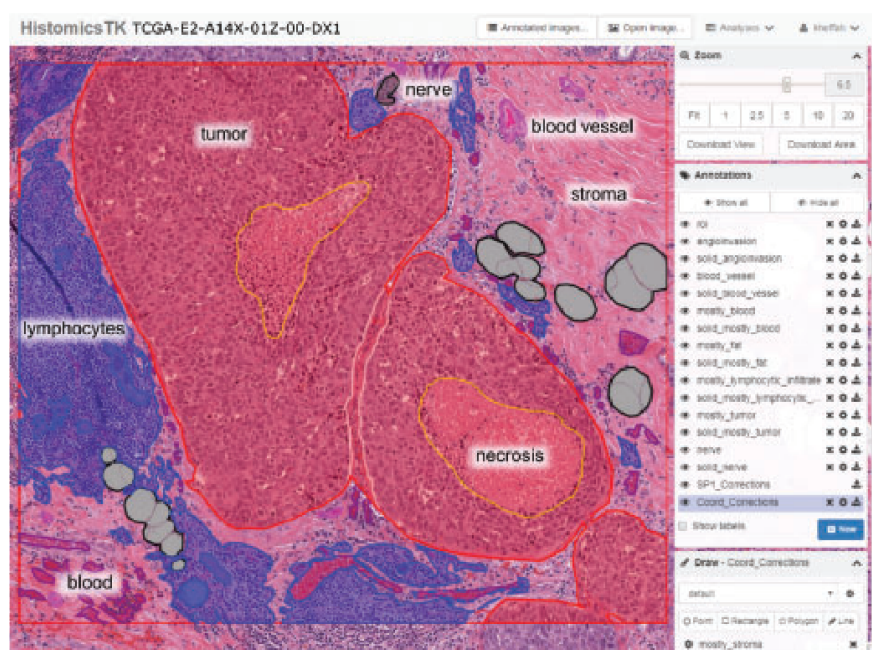


Figure 2-9 Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [Amgad et al., 2019].

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CHAPTER

THREE

1167

CHAINED GAUSSIAN PROCESSES

1168

3.1 Background and Related Methods

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In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators [Yan et al., 2014] [Zhang, 2022]. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

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3.1.1 Kernel Alignment-Based Annotator Relevance Analysis

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The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [Gil-Gonzalez et al., 2018], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is unavailable. The key

1179 innovation of **KAAR** lies in its use of **Centered Kernel Alignment (CKA)** to measure
 1180 the similarity between input features and annotator labels.

1181 Given a dataset with input features $\mathbf{X}$ and labels from multiple annotators $\mathbf{Y}$, **KAAR**
 1182 computes a kernel matrix $\mathbf{K}_\nu$ as a convex combination of R basis kernels:

$$\mathbf{K}_\nu = \sum_{r=1}^R \nu_r \mathbf{K}_r, \quad (3-1)$$

1183 where $\mathbf{K}_r$ is computed using a kernel function over the input features, and ν_r
 1184 represents the relevance weight for the r -th annotator. The weights are optimized
 1185 by maximizing the CKA between $\mathbf{K}_\nu$ and a target kernel $\mathbf{F}$ computed from the
 1186 input features:

$$\rho(\mathbf{K}_\nu, \mathbf{F}) = \frac{\langle \bar{\mathbf{K}}_\nu, \bar{\mathbf{F}} \rangle_{\text{F}}}{\|\bar{\mathbf{K}}_\nu\|_{\text{F}} \|\bar{\mathbf{F}}\|_{\text{F}}}, \quad (3-2)$$

1187 A key advantage of **KAAR** is its ability to capture dependencies between annotators
 1188 through the kernel framework, without requiring explicit parametric modeling of
 1189 these relationships.

1190 **3.1.2 Localized Kernel Alignment-Based Annotator** 1191 **Relevance Analysis**

1192 In [Gil-Gonzalez et al., 2021a] the authors extend **KAAR** by introducing locality in
 1193 the kernel alignment framework. Rather than assuming constant annotator
 1194 performance across the input space, **Localized Kernel Alignment-Based Annotator**
 1195 **Relevance Analysis (LKAAR)** models the annotator relevance as a function of the
 1196 input features. The kernel combination in **LKAAR** takes the form:

$$\mathbf{K}_q = \sum_{r=1}^R \mathbf{Q}_r \mathbf{K}_r \mathbf{Q}_r, \quad (3-3)$$

1197 where $\mathbf{Q}_r$ is a diagonal matrix whose elements represent the local relevance of
 1198 annotator r . These elements are computed as:

$$q_r(\mathbf{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^N \beta_{n'}^{(r)} \kappa_\beta(\mathbf{x}_n, \mathbf{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \quad (3-4)$$

1199 This localized approach allows **LKAAR** to better model inconsistent annotators
 1200 whose performance varies across different regions of the input space [**Gil-Gonzalez**
 1201 **et al., 2021a**].

1202 3.1.3 Chained Gaussian Processes Model

1203 The **CGP** model provides a probabilistic framework for modeling multiple outputs
 1204 through a chain of Gaussian processes [**Saul et al., 2016**]. In the context of multiple
 1205 annotators, **CGP** can be used to model both the ground truth and annotator
 1206 behavior. The key idea is to model a likelihood function with J parameters as in
 1207 [**Giraldo et al., 2022**]:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{n=1}^N p(y_n | \theta_1(\mathbf{x}_n), \dots, \theta_J(\mathbf{x}_n)), \quad (3-5)$$

1208 where each parameter $\theta_j(\mathbf{x})$ is modeled using a separate Gaussian process. This
 1209 framework provides remarkable flexibility in modeling complex relationships
 1210 within the data [**Giraldo et al., 2022**]. The use of Gaussian processes enables
 1211 natural uncertainty quantification through posterior distributions, allowing the
 1212 model to express varying degrees of confidence in its predictions. The framework

1213 inherently handles missing data through its probabilistic nature, and the automatic
 1214 relevance determination property of GPs helps identify the most important
 1215 features for prediction [Giraldo et al., 2022].

1216 The CGP framework becomes particularly powerful when integrated with concepts
 1217 from KAAR and LKAAR. By incorporating kernel alignment principles, the model
 1218 can make more informed choices about kernel functions, better capturing the
 1219 underlying structure of the data [Giraldo et al., 2022]. The framework can be
 1220 extended to include input-dependent lengthscales, allowing for varying degrees of
 1221 smoothness across the input space [Giraldo et al., 2022]. Furthermore, through
 1222 multi-output Gaussian Process (GP) formulations, the model can capture complex
 1223 dependencies capture complex dependencies between different annotators,
 1224 providing a rich representation of how different experts' opinions relate to each
 1225 other.

1226 However, implementing CGPs in practice presents several significant challenges.
 1227 The computational complexity of the model scales cubically with the dataset size,
 1228 making it potentially prohibitive for large-scale applications [Liu et al., 2020]. The
 1229 need for approximate inference techniques in non-Gaussian likelihood scenarios
 1230 can introduce additional complexity and potential approximation errors
 1231 [Rasmussen and Williams, 2006a]. When dealing with high-dimensional inputs,
 1232 the model may struggle with the curse of dimensionality, requiring careful feature
 1233 selection or dimensionality reduction preprocessing [Rasmussen and Williams,
 1234 2006a]. Additionally, the selection of appropriate kernel functions remains a
 1235 critical challenge, as it significantly impacts the model's performance and its ability
 1236 to capture relevant patterns in the data [Rasmussen and Williams, 2006b].

1237 3.1.4 Regularized Chained Deep Neural Network

1238 In [Gil-Gonzalez et al., 2021b], the authors propose a deep learning approach to
 1239 multiple annotator learning, inspired by the chained Gaussian processes model. The

method models both the ground truth estimation and annotator reliability through a deep neural network architecture. The likelihood function in **Regularized Chained Deep Neural Network (RCDNN)** takes the form:

$$p(\mathbf{Y}|\boldsymbol{\theta}) = \prod_{n=1}^N \prod_{r \in R_n} \left(\prod_{k=1}^K \zeta_{n,k}^{\delta(y_n^{(r)} - k)} \right)^{\lambda_n^{(r)}} \left(\frac{1}{K} \right)^{1 - \lambda_n^{(r)}}, \quad (3-6)$$

where $\lambda_n^{(r)}$ represents the reliability of annotator r for instance n , and $\zeta_{n,k}$ is the estimated probability of class k for instance n .

The **RCDNN** architecture is built upon a sophisticated multi-layer structure that combines several key innovations in deep learning [Gil-Gonzalez et al., 2021b]. At its core, it employs multiple dense layers with carefully chosen activation functions, each designed to capture different aspects of the annotator-data relationship. To prevent overfitting, a comprehensive regularization strategy is implemented, combining both L1 and L2 norms with dropout mechanisms [Gil-Gonzalez et al., 2021b]. The network architecture splits into specialized branches: one dedicated to estimating the ground truth labels, and another focused on modeling annotator reliability. This dual-branch approach allows the network to simultaneously learn both the true underlying patterns in the data and the individual characteristics of each annotator. Furthermore, the implementation of Monte Carlo dropout during prediction provides a mechanism for uncertainty estimation, offering not just predictions but also confidence measures for those predictions.

3.2 Tooling and Implementation

The implementation of Chained Gaussian Processes for multiple annotator learning requires robust and flexible software tools that can handle both the probabilistic

1262 nature of Gaussian processes and the complexity of deep architectures [Saul et al.,
1263 2016]. In this work, two key frameworks are leveraged: GPflow and GPflux. These
1264 tools provide the foundation for building scalable and efficient implementations of
1265 our models while maintaining the theoretical rigor necessary for proper uncertainty
1266 quantification.

1267 The choice of these frameworks is motivated by several factors. First, both are
1268 built on top of TensorFlow, providing access to automatic differentiation and GPU
1269 acceleration capabilities. Second, they offer a modular design that allows for easy
1270 extension and modification, which is crucial when implementing novel model
1271 architectures. Third, they provide robust implementations of various Gaussian
1272 process models and inference methods, allowing us to focus on the specific
1273 challenges of multiple annotator learning rather than reimplementing basic GP
1274 functionality.

1275 3.2.1 GPflow Overview

1276 GPflow is a Python package for building Gaussian process models in TensorFlow.
1277 Developed as a modern and scalable alternative to GPy, it leverages TensorFlow’s
1278 computational capabilities to provide efficient implementation of Gaussian process
1279 models. The framework is designed with four key principles in mind: ease of use,
1280 computational efficiency, extensibility, and automatic differentiation. This work
1281 was originally published in [de G. Matthews et al., 2016].

1282 At its core, GPflow provides implementations of various Gaussian process models
1283 including Gaussian Process Regression (GPR), Gaussian Process Monte Carlo
1284 (GPMC), Sparse Gaussian Process Regression (SGPR), and Sparse Gaussian Process
1285 Monte Carlo (SGPMC) [de G. Matthews et al., 2016]. These implementations are
1286 built on top of TensorFlow’s computational graph framework, allowing for
1287 automatic differentiation and GPU acceleration where available.

1288 The framework’s architecture is modular, with clear separation between model
1289 components such as kernels, likelihoods, and optimization routines. This
1290 modularity makes it straightforward to extend existing models or create new ones.
1291 GPflow includes a comprehensive suite of kernel functions, ranging from the basic
1292 RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral
1293 Mixture kernel.

1294 One of GPflow’s distinguishing features is its robust handling of model parameters.
1295 It provides facilities for parameter transformation (ensuring parameters remain in
1296 valid ranges during optimization), prior specification, and flexible optimization
1297 strategies. The framework also includes utilities for model evaluation and
1298 prediction, with built-in support for uncertainty estimation.

1299 **3.2.2 GPflux Framework**

1300 GPflux extends GPflow to enable the construction and training of Deep Gaussian
1301 Process (DGP) models using modern deep learning practices. Built on top of both
1302 GPflow and TensorFlow, GPflux provides a flexible framework for implementing
1303 deep probabilistic models that combine the expressiveness of deep neural
1304 networks with the uncertainty quantification capabilities of Gaussian processes.
1305 This work was originally published in [Dutordoir et al., 2021].

1306 The framework implements Deep Gaussian Processes as a composition of Gaussian
1307 process layers, where each layer transforms its inputs through a Gaussian process
1308 mapping. This hierarchical structure allows the model to capture complex,
1309 non-stationary patterns in data while maintaining uncertainty estimates
1310 throughout the entire processing pipeline. GPflux handles this complexity through
1311 a carefully designed architecture that separates the concerns of model
1312 specification, inference, and prediction.

1313 A key innovation in GPflux is its integration with Keras-style layers, allowing
1314 practitioners to combine deterministic neural network layers with Gaussian

process layers in a single model. This hybrid approach enables the construction of models that can leverage both the structural assumptions encoded in neural network architectures and the flexibility of Gaussian processes.

For inference, GPflux implements various approximate methods, including stochastic variational inference with inducing points, which makes it possible to train deep Gaussian process models on large datasets. The framework provides built-in support for minibatch training, making it practical to work with datasets that don't fit in memory.

3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [Gil-Gonzalez et al., 2023], which are as follows:

3.3.1 Classification

A fully synthetic data for a 1-D ($P = 1$) multiclass classification problem with $K = 3$ classes is generated. The input feature matrix $\mathbf{X}$ is constructed by randomly sampling $N = 100$ points from a uniform distribution over the interval $[0, 1]$. For each sample n , the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$

and assigning the class label as the index $i \in \{1, 2, 3\}$ that maximizes $t_{n,i}$. This construction ensures a nontrivial, nonlinear decision boundary between the classes [Gil-González and Álvarez Meza, 2023].

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval $[0, 1]$, providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an SLFM-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q : \mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$, where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D representation of the input features in $\mathbf{X}$, obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.
3. **Reliability calculation:** Calculate $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$, where $\varsigma(\cdot) \in [0, 1]$ is the inverse exponential sigmoid function.

4. **Label assignment:** The r th annotator's label for the n th sample is given by

$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$

where y_n is the true label and $\tilde{y}_n$ is a flipped version of the actual label y_n .

This procedure ensures that the simulated annotator labels exhibit both inter-annotator dependencies and input-dependent reliability.

1357 3.3.2 Regression

1358 A fully synthetic dataset is also generated for a 1-D regression problem, where the
1359 ground truth for the n th sample is defined as:

$$y_n = \sin(2\pi x_n) \sin(6\pi x_n) \quad (3-8)$$

1360 Here, the input matrix $\mathbf{X}$ is constructed by randomly sampling 100 points uniformly
1361 from the interval $[0, 1]$. The test set is created by extracting equally spaced samples
1362 from the same interval, ensuring a clear separation between training and testing
1363 data.

1364 To simulate the presence of multiple annotators, the ground truth is corrupted with
1365 annotator-specific noise. For each annotator r , the observed label for the n th sample
1366 is given by:

$$y_n^r = y_n + \epsilon_n^r$$

1367 where $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$ is Gaussian noise, and v_n^r represents the error variance for
1368 the r th annotator, which may depend on the input features. This setup allows us
1369 to model both the bias and the reliability of each annotator, as well as potential
1370 dependencies between annotators.

1371 To further increase realism, the error variance v_n^r can be modeled as a function of
1372 the input features, capturing scenarios where annotator reliability varies across the
1373 input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

1374 being v_n^r , the r th annotator's error variance for the n th sample.

1375 Again, for modeling the error as a function of the input features, an **SLFM**-based
1376 approach (termed SLFM-R) is employed to build the noisy labels, as follows:

- 1377 **1. Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q :$
 1378 $\mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
- 1379 **2. Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$,
 1380 where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D
 1381 representation of the input features in $\mathbf{X}$, obtained using the t-distributed
 1382 Stochastic Neighbor Embedding (t-SNE) approach.
- 1383 **3. Compute variance :** Finally, determine $v_n^r = \exp(\hat{f}_{l,r,n})$.

1384 3.4 Results

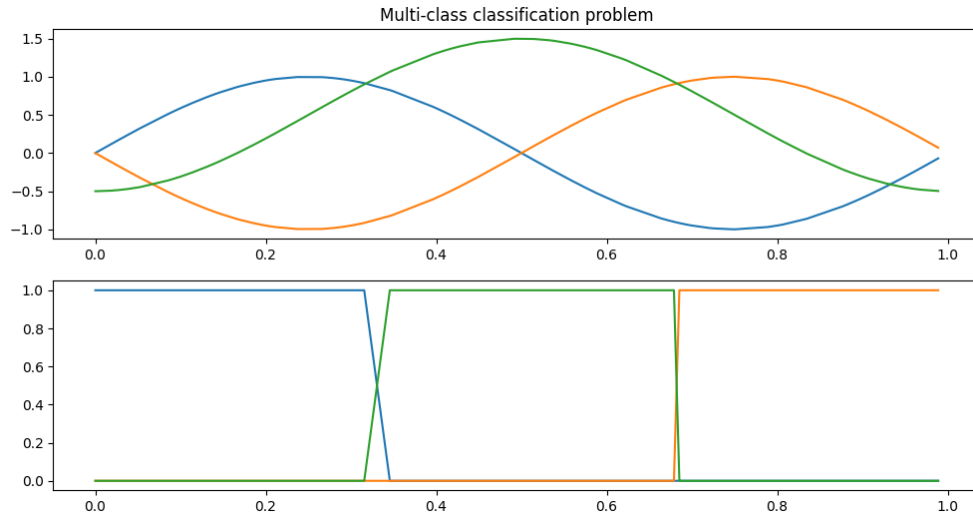
1385 The built synthetic datasets are used to train the **CCGPMA** model. Emulated
 1386 classification labels following the true label determining function from 3-7 can be
 1387 seen in Figure 3-1. On the other hand, the ground truth and noisy labels emulation
 1388 for annotators across the feature space for regression task can be seen in Figure
 1389 3-3.

1390 Using both GPflux and GPflow for the regression and classifications tasks, leads to
 1391 the following combinations experiments: GPflux-CCGPMA Regression,
 1392 GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification,
 1393 GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification,
 1394 GPflux-CCGPMA Multiclass Classification. All of the experiments converged
 1395 successfully with almost identical results and close to no difference in
 1396 performance.

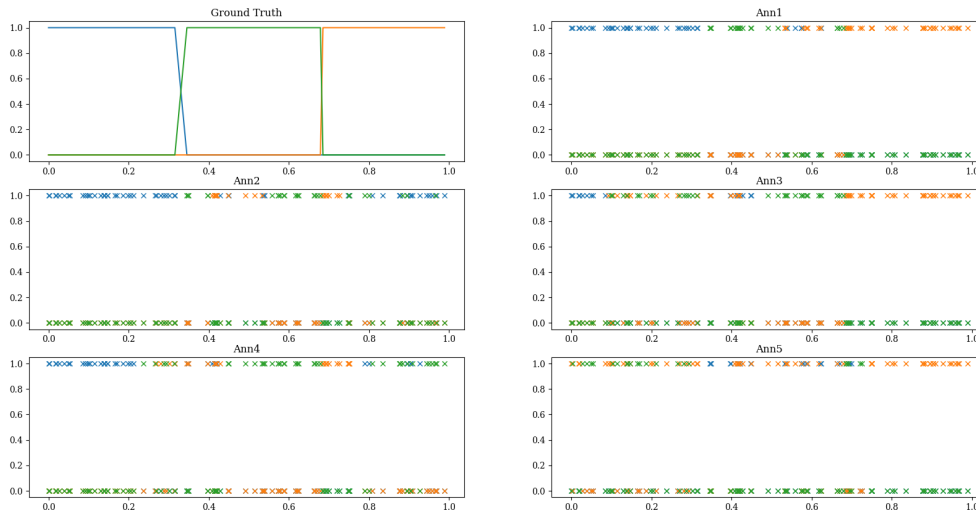
1397 3.5 Conclusion

1398 3.5.1 Comparative Analysis

1399 The evolution of multiple annotator learning methods represents a fascinating
 1400 progression in how we approach the challenge of learning from diverse expert



(a) Ground truth labels across the feature space



(b) Noisy emulations for annotators across the feature space for classification task

Figure 3-1 Synthetic dataset overview for classification task

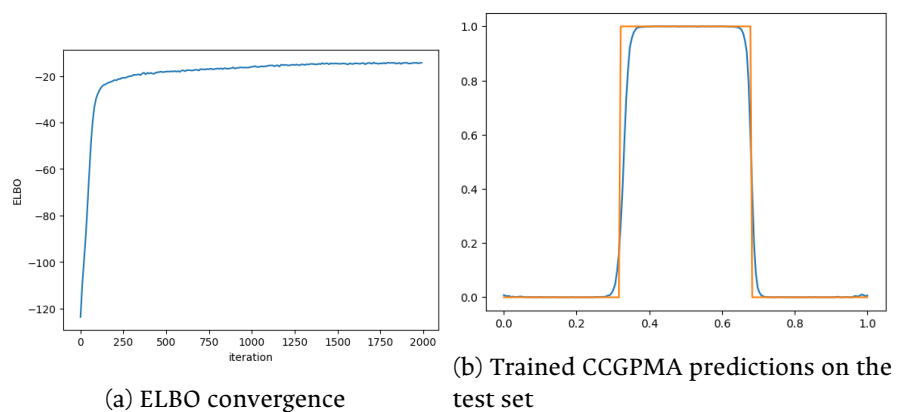


Figure 3-2 Classification results for the CCGPMA model using GPflux.

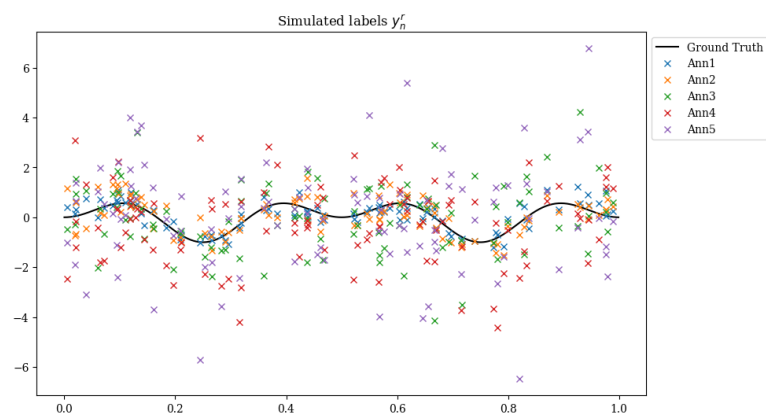
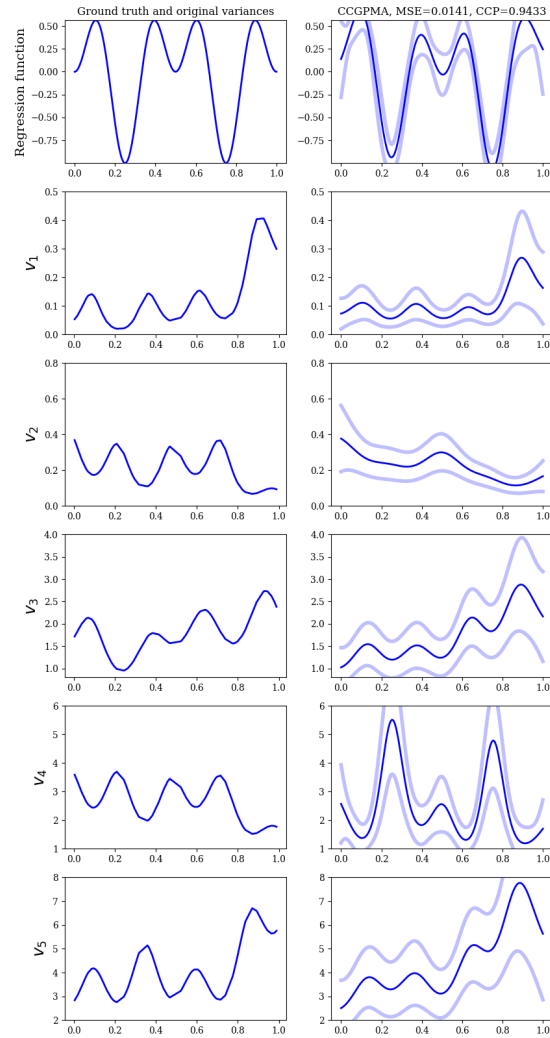


Figure 3-3 Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure 3-4 Regression results for the CCGPMA model using GPflux.

opinions. **KAAR** established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. **LKAAR** advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. **RCDNN** brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. **CCGPMA** represents perhaps the most theoretically elegant approach, providing a principled probabilistic framework, though it faces significant computational challenges in practical applications.

The selection of an appropriate method for a given application requires careful consideration of several factors. The size and dimensionality of the dataset play a crucial role, as some methods scale better than others with increasing data volume. The need for uncertainty quantification may favor probabilistic approaches like CGP, while computational constraints might push towards more efficient methods like KAAR. The presence of input-dependent annotator behavior could make **LKAAR** or **RCDNN** more appropriate choices. Additionally, when interpretability is a key requirement, as in many medical applications, methods like **KAAR** and **LKAAR** might be preferred over the more complex Deep Learning approaches. Table 3-1 provides a comprehensive comparison of the key characteristics and capabilities of each method discussed.

Table 3-1 Comparison of Multiple Annotator Learning Methods

Characteristic	KAAR [Gil-Gonzalez et al., 2018]	LKAAR [Gil-Gonzalez et al., 2021a]	RCDNN [Gil-Gonzalez et al., 2021b]	CGP [Gil-González and Álvarez Meza, 2023]
Input-dependent annotator behavior	No	Yes	Yes	Yes
Annotator dependencies modeling	Yes	Yes	Yes	Yes
Uncertainty quantification	No	No	Partial	Yes
Computational scalability	High	Medium	High	Low
Missing data handling	Yes	Yes	Yes	Yes
Non-linear relationships	Yes	Yes	Yes	Yes
Training complexity	Low	Medium	High	High
Hyperparameter sensitivity	Low	Medium	High	Medium
Interpretability	High	Medium	Low	Medium
Memory requirements	Low	Medium	High	High

Note: Partial uncertainty quantification in **RCDNN** refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

1422 3.5.2 Implementation Considerations

1423 When implementing Chained Gaussian Processes for multiple annotator learning
 1424 using these tools, several key considerations must be addressed. First, the model
 1425 architecture must be carefully designed to capture both the individual annotator
 1426 characteristics and their interdependencies [Gil-González and Álvarez Meza, 2023].
 1427 This requires extending the basic GP layers provided by GPflux to handle multiple
 1428 output streams - one for each annotator’s reliability model.

1429 Second, the inference procedure must be adapted to handle the unique challenges
 1430 of multiple annotator data, such as missing labels and varying levels of annotator
 1431 expertise across different regions of the input space. The variational inference
 1432 capabilities of GPflux provide a solid foundation for this, but careful attention
 1433 must be paid to the design of the variational approximation to ensure it captures
 1434 the relevant uncertainty in both the ground truth and annotator reliability
 1435 estimates.

1436 Third, kernel selection and design presents a critical challenge in implementing the
 1437 CGP model. The choice of kernel functions significantly impacts the model’s ability
 1438 to capture the underlying structure of the data and the relationships between
 1439 annotators [Rasmussen and Williams, 2006b]. Several considerations must be
 1440 taken into account:

- 1441 • **Kernel Complexity:** The model requires multiple kernel functions to capture
 1442 different aspects of the data - one for the ground truth and additional kernels
 1443 for each annotator’s reliability model. This multi-kernel setup increases the
 1444 complexity of kernel selection and optimization.
- 1445 • **Input-Dependent Behavior:** Since annotator performance often varies across
 1446 the input space, the kernel functions must be flexible enough to capture this
 1447 non-stationary behavior. This may require using composite kernels or kernels
 1448 with input-dependent lengthscales.

- 1449 • **Inter-Annotator Dependencies:** The kernel design must account for potential
1450 correlations between different annotators' behaviors. This can be achieved
1451 through careful design of the kernel structure in the SLFM framework, where
1452 the combination of basis kernels must be chosen to effectively model these
1453 dependencies.
- 1454 • **Computational Efficiency:** The choice of kernel functions also impacts
1455 computational performance. While more expressive kernels may better
1456 capture complex patterns, they often come with increased computational
1457 cost. This trade-off must be carefully considered, especially when dealing
1458 with large datasets.

1459 Finally, computational efficiency becomes a critical concern when scaling to large
1460 datasets with multiple annotators. The minibatch training capabilities of both
1461 frameworks help address this, but additional considerations such as appropriate
1462 inducing point selection and batch size tuning become important for achieving
1463 good performance in practice.

1464 3.5.3 Final Remarks

1465 In this chapter, we have presented a comprehensive overview of the Chained
1466 Gaussian Processes model and its application to multiple annotator learning. We
1467 have discussed the theoretical underpinnings of the model, its implementation
1468 using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in
1469 [Gil-Gonzalez et al., 2023] that CCGPMA outperforms other method in state of the
1470 art methods in terms of accuracy and uncertainty quantification (see 1.3.1).

1471 While Gaussian Processes have demonstrated remarkable success in regression
1472 and classification tasks, their application to image segmentation presents
1473 significant computational challenges [Rasmussen and Williams, 2006b]. The cubic
1474 computational complexity of GPs ($\mathcal{O}(N^3)$), which scales with the number of data

1475 points, becomes particularly problematic when dealing with high-resolution
1476 images where each pixel can be considered a data point [Liu et al., 2020]. This
1477 limitation makes traditional GP approaches impractical for large-scale image
1478 segmentation tasks, where the number of pixels can easily reach into the millions.
1479 Alternative approaches, such as deep learning-based segmentation methods, often
1480 provide more computationally efficient solutions for these high-dimensional
1481 problems, though at the cost of uncertainty quantification capabilities that GPs
1482 naturally provide [Kendall and Gal, 2017] [Wilson et al., 2016].

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CHAPTER

FOUR

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1488

TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

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4.1 Loss Functions for Multiple Annotators

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As mentioned in Section 2.3.2, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

1497 4.1.1 Generalized Cross Entropy

1498 The Generalized Cross Entropy (GCE) loss function was first introduced by [Zhang
1499 and Sabuncu, 2018] as a robust alternative to the standard cross-entropy loss,
1500 particularly effective in handling noisy labels. Let us first consider the Cross
1501 Entropy (CE) and Mean Absolute Error (MAE) loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \quad (4-1)$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^K y_k \log(f_k(\mathbf{x})) \quad (4-2)$$

1502 where $y_k \in \mathbf{y}$, $f_k(\mathbf{x}) \in f(\mathbf{x})$, and $\|\cdot\|_1$ stands for the l_1 -norm. Of note, $\mathbf{1}^\top \mathbf{y} =$
1503 $\mathbf{1}^\top f(\mathbf{x}) = 1$, $\mathbf{1} \in \{1\}^K$ being an all-ones vector. In addition, the MAE loss can be
1504 rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{y}^\top \odot f(\mathbf{x})) \quad (4-3)$$

1505 where $\odot$ stands for the element-wise product.

1506 The CE loss function exhibits several distinct characteristics that make it
1507 particularly sensitive to noisy labels. It is unbounded from above and heavily
1508 penalizes confident but wrong predictions, which can lead to instability in the
1509 presence of label noise. In contrast, the MAE loss offers a more robust alternative
1510 with its bounded nature and symmetric properties in softmax-based
1511 representations. While the MAE loss is more resilient to noisy labels, it tends to
1512 train more slowly due to its equal weighting of all mistakes regardless of
1513 confidence.

1514 The GCE loss function, as defined by the authors in [Zhang and Sabuncu, 2018],
1515 provides a flexible framework that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2 \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q}, \quad (4-4)$$

with $q \in (0, 1]$. Remarkably, the limiting case for $q \rightarrow 0$ in GCE is equivalent to the CE expression, and when $q = 1$, it equals the MAE loss. In addition, the GCE holds the following gradient with regard to θ :

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x}; \theta)|_k)}{\partial \theta} = -f_k(\mathbf{x}; \theta)^{q-1} \nabla_\theta f_k(\mathbf{x}; \theta). \quad (4-5)$$

The GCE loss combines the best of both worlds, offering robustness to label noise while maintaining the convexity property necessary for optimization. The truncation parameter q provides a mechanism to control the sensitivity to outliers, allowing for fine-tuning of the loss function's behavior based on the specific characteristics of the dataset.

4.1.2 Extension to Multiple Annotators

In the context of multiple annotators, we need to consider the varying reliability of each annotator across different regions of the image. Let's consider a k -class multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W \times H}, \{\mathbf{Y}_r \in \{0, 1\}^{W \times H \times K}\}_{r=1}^R; \hat{\mathbf{Y}} \in [0, 1]^{W \times H \times K} = \mathbf{f}(\mathbf{X}) \quad (4-6)$$

where the segmentation mask function maps the input to output as:

$$\mathbf{f}: \mathbb{R}^{W \times H} \rightarrow [0, 1]^{W \times H \times K} \quad (4-7)$$

The segmentation masks $\mathbf{Y}_r$ satisfy the following condition for being a softmax-like representation:

$$\mathbf{Y}_r[w, h, :] \mathbf{1}_k^\top = 1; \quad w \in W, h \in H \quad (4-8)$$

1531 4.1.3 Reliability Maps and Truncated GCE

1532 The key innovation in our approach is the introduction of reliability maps Λ_r for
1533 each annotator:

$$\left\{ \Lambda_r(\mathbf{X}; \theta) \in [0, 1]^{W \times H} \right\}_{r=1}^R \quad (4-9)$$

1534 These reliability maps serve as a mechanism to estimate the confidence of each
1535 annotator at every spatial location (w, h) in the image. By learning these maps
1536 jointly with the segmentation model, the network gains the ability to adapt to the
1537 varying levels of expertise across different regions of the image. This approach
1538 allows for dynamic weighting of each annotator's contribution based on their
1539 estimated reliability in specific areas, effectively handling cases where annotators
1540 might demonstrate different levels of expertise in different parts of the image.

1541 The proposed Truncated Generalized Cross Entropy for Semantic Segmentation
1542 (TGCE_{SS}) combines the robustness of GCE with the flexibility of reliability maps.
1543 Let us first define the key components:

$$\mathcal{L}_{GCE}(\mathbf{Y}_r, f(\mathbf{X}; \theta)) = \mathbb{E}_k \left\{ \mathbf{Y}_r \odot \left(\frac{\mathbf{1}_{W \times H \times K} - f(\mathbf{X}; \theta)^{\odot q}}{q} \right) \right\} \quad (4-10)$$

$$\mathcal{L}_{prior} = \frac{\mathbf{1}_{W \times H} - \left(\frac{1}{K} \mathbf{1}_{W \times H} \right)^{\odot q}}{q} \quad (4-11)$$

1544 where $\mathcal{L}_{GCE}$ represents the GCE loss component and $\mathcal{L}_{prior}$ represents the uniform
1545 prior component. The complete TGCE_{SS} loss function is then defined as:

$$\begin{aligned} TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X}; \theta) |_{r(\mathbf{X}; \theta)}) = \mathbb{E}_{r,w,h} \left\{ \right. \\ \Lambda_r(\mathbf{X}; \theta) \odot \mathcal{L}_{GCE} + \tilde{C}(\mathbf{1}_{W \times H} - \Lambda_r(\mathbf{X}; \theta)) \odot \mathcal{L}_{prior}; \\ \left. w \in W, h \in H; r \in R \right\} \end{aligned} \quad (4-12)$$

1546 where $q \in (0, 1)^1$ controls the MAE or CE level in the same way as in the GCE loss
 1547 function. $\tilde{C}$ controls is an additional hyperparameter which controls the tolerance
 1548 to unreliable annotations. The loss function consists of two main components:

- 1549 • The first term weighted by Λ_r represents the GCE loss for regions where the
 1550 annotator is considered reliable
- 1551 • The second term weighted by $(1 - \Lambda_r)$ and $\tilde{C}$ provides a uniform prior for
 1552 regions where the annotator is considered unreliable

1553 A summary of the loss function mechanism is shown in Figure 4-1.

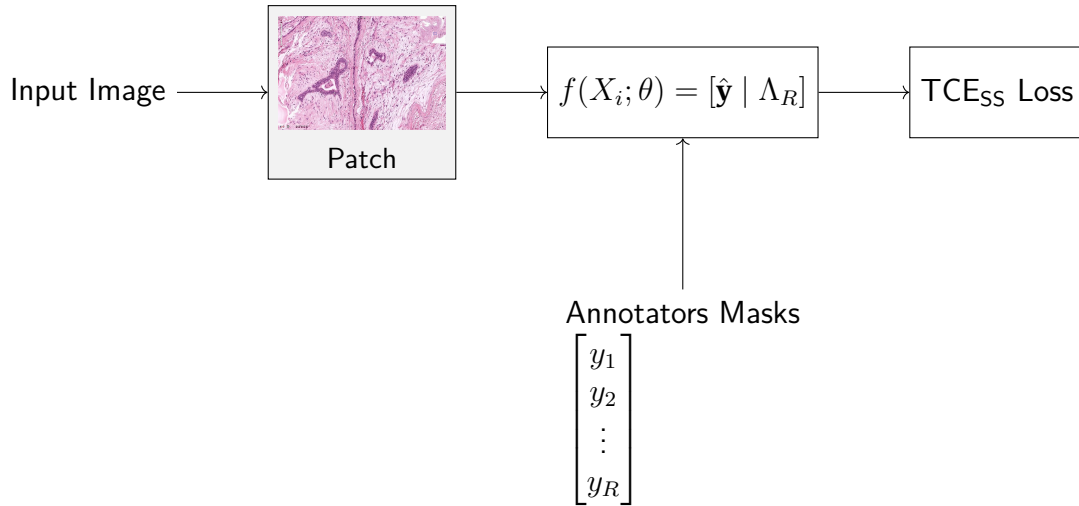


Figure 4-1 Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The loss combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$ to compute a weighted loss that accounts for annotator reliability across different image regions.

¹To ensure numerical stability in the loss function computations, particularly in division operations, a small epsilon value ϵ is added to denominators in practical applications. This prevents division by zero errors and stabilizes gradient calculations during training. The epsilon value is kept small enough (10^{-6}) to not significantly impact the loss function behavior while providing the necessary numerical safeguards.

1554 4.2 Chapter overview

1555 This chapter presents a novel approach to handle multiple annotators in semantic
 1556 segmentation tasks through the introduction of the Truncated Generalized Cross
 1557 Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The key contributions
 1558 and novelties of this work can be summarized as follows:

- 1559 • **Reliability-aware Learning:** Unlike traditional approaches that treat all
 1560 annotators equally, the proposed method introduces reliability maps (Λ_r)
 1561 that dynamically estimate the confidence of each annotator at every spatial
 1562 location in the image. This allows the model to adapt to varying levels of
 1563 expertise across different regions.
- 1564 • **Robust Loss Function:** The proposed TGCE_{SS} combines the benefits of both
 1565 Cross Entropy and Mean Absolute Error losses through the Generalized
 1566 Cross Entropy framework. The truncation parameter q provides a flexible
 1567 mechanism to control the sensitivity to noisy labels while maintaining the
 1568 convexity property necessary for optimization.
- 1569 • **Unified Framework:** The approach provides a unified framework that
 1570 handles both reliable and unreliable regions through a weighted
 1571 combination of GCE loss and uniform prior. This is particularly important in
 1572 medical imaging where annotator expertise can vary significantly across
 1573 different anatomical structures.

1574 The main novelty of this approach lies in its ability to simultaneously learn the
 1575 segmentation task while estimating annotator reliability in a spatially-aware
 1576 manner. This represents a significant advancement over existing methods that
 1577 either:

- 1578 • Treat all annotators equally without considering their varying expertise
- 1579 • Use global reliability scores that don't account for spatial variations in
 1580 annotator performance

- 1581 • Rely on fixed, pre-defined reliability measures rather than learning them from
1582 data

1583 The proposed $TGCE_{SS}$ loss function represents a step forward in handling the
1584 complex nature of multiple annotator scenarios in semantic segmentation,
1585 particularly in medical imaging applications where expert knowledge can vary
1586 significantly across different regions of interest. This work provides a foundation
1587 for more sophisticated approaches to handling multiple annotators in deep
1588 learning-based segmentation tasks.

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CHAPTER

FIVE

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CHAINED DEEP LEARNING AND REGULARIZED $TGCE_{SS}$

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LOSS FOR IMAGE SEGMENTATION

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As mentioned in Chapter 1, U-shaped Convolutional Neural Networks (CNNs) have been proven a good solution for Semantic segmentation (SS) tasks in medical images, due to their ability to capture both global and local information with small datasets.

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The proposed model architecture combines the strengths of U-Net with a ResNet-34 backbone, specifically designed to work with the $TGCE_{SS}$ loss function by providing a multi-output network which focuses on two tasks: **Multiclass Semantic Segmentation** and **Reliability Estimation**. The architecture is illustrated in Figure 5-2.

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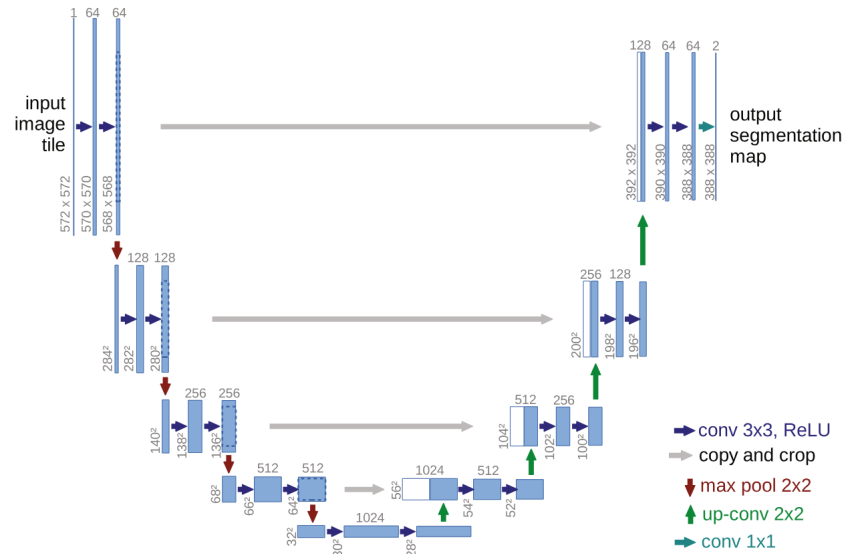


Figure 5-1 Original U-NET architecture from [Ronneberger et al., 2015].

5.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training [He et al., 2015]. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

5.1.1 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The

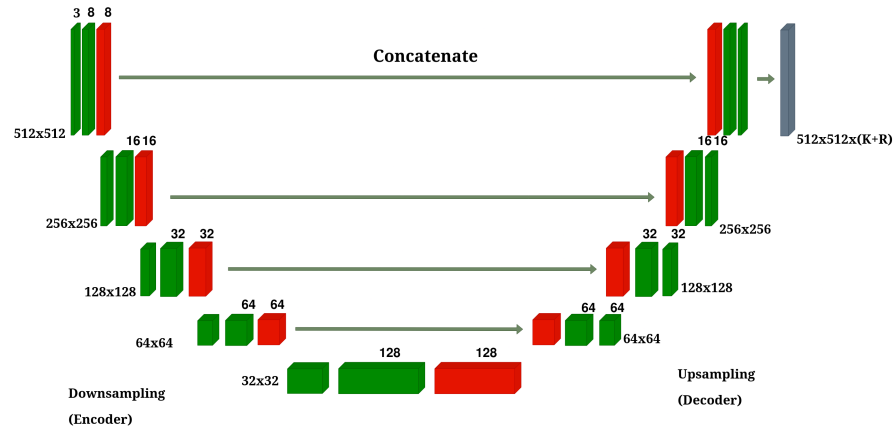


Figure 5-2 Overview of the proposed model architecture.

decoder path employs transposed convolutions for upsampling, creating a symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

5.2 Reliability Branch

5.2.1 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a

multi-output network. The final output consists of R reliability maps Λ_r , one for each annotator, with values constrained to the $[0, 1]$ range through a sigmoid activation function. This design allows the network to learn and adapt to the varying reliability of different annotators across different regions of the image.

The scalar reliability branch is implemented by taking the bottleneck features from the encoder and applying a global average pooling operation followed by a dense layer with sigmoid activation. This results in a single reliability value per annotator for each input image, making it computationally efficient but less spatially aware of reliability variations within the image.

5.2.2 Feature-based Reliability Estimation

The feature-based reliability estimation approach provides a more detailed spatial understanding of annotator reliability by operating on the bottleneck features of the encoder. This approach maintains the spatial information from the encoder's bottleneck while being more computationally efficient than pixel-wise reliability estimation.

The feature-based reliability branch is implemented by applying two convolutional layers with batch normalization to the bottleneck features:

- A first convolutional layer with 32 filters to process the bottleneck features
- A second convolutional layer that produces R feature maps, one for each annotator

This approach provides a coarse-grained spatial reliability map that captures regional variations in annotator reliability while maintaining computational efficiency. The resulting reliability maps have the same spatial dimensions as the bottleneck features, providing a middle ground between scalar and pixel-wise reliability estimation.

5.2.3 Pixel-wise Reliability Estimation

The pixel-wise reliability estimation approach provides the most detailed spatial understanding of annotator reliability by operating on the full-resolution features from the decoder. This approach allows for fine-grained reliability estimation at the pixel level, capturing local variations in annotator reliability.

The pixel-wise reliability branch is implemented by applying three convolutional layers with batch normalization to the decoder's output features:

- Two initial convolutional layers with 32 filters each to process the features
- A final convolutional layer that produces R reliability maps, one for each annotator

This approach provides the most detailed spatial reliability information, allowing the model to capture pixel-level variations in annotator reliability. The resulting reliability maps have the same spatial dimensions as the input image, providing the finest granularity of reliability estimation among the three approaches.

5.3 Applying regularization terms to the $TGCE_{SS}$ loss function

The model produces two distinct outputs: segmentation masks $\mathbf{\hat{Y}} = f(\mathbf{X}; \theta)$ and reliability maps $\{\Lambda_r(\mathbf{X}; \theta)\}_{r=1}^R$. These outputs work in tandem with the $TGCE_{SS}$ loss function described in Section 4.1. The loss function simultaneously guides the learning of both the segmentation masks and reliability maps, ensuring that the model learns to balance the contributions of different annotators based on their estimated reliability.

The $TGCE_{SS}$ loss function is implemented differently for each reliability approach, with specific adaptations to handle the different spatial dimensions of the

reliability maps. Naturally, the model itself contains tunable hyperparameters related to the Generalized Cross Entropy (GCE) - Mean Absolute Error (MAE) trade off behavior (also known as q parameter) and the unreliability tolerance $\tilde{C}$, however, some additional hyperparameters were added with the purpose of regularizing the overall loss for every experiment.

These additional regularization terms arise from the need to ensure consistency in the internal representations of the model, since some unregularized experiments showed that the model could sometimes converge to low loss values but outputs in the reliability branch with no consistency at all, since the model might assign extreme reliability values to some labelers and not others. The added regularization terms help the model converge to reliability values that are more consistent across all labelers, that do not reach extreme values and that maintain high reliability scores for scorers that did apply a label in the first place and also were considered better labelers ¹

5.3.1 Centroid penalty for extreme reliabilities

Some optimization processes might favor a convergence to extreme reliability estimations, by assigning either 0 or 1 to all labelers. To avoid this, a regularization term is added to the loss function which penalizes the model for assigning extreme reliability values and favoring those around a centroid. High scaling values for this regularization term might converge to a very homogeneous reliability map across all labelers, which is not desired either.

$$\mathcal{L}_{centroid} = \alpha_{centroid} \cdot \mathbb{E}_{r \in R}[(\Lambda_r - 0.5)^2] \quad (5-1)$$

The behavior of this regularization term is shown in Figure 5-3 (a).

¹For this, the Oxford PET dataset with emulated noisy labelers was very useful since the labeler reliability (related to noise levels in encoder layers) was very easy to control, making it possible to debug the model and note if reliability estimations levels were consistent with the labelers considered better. This was the main criteria to design and select the regularization terms.

1702 5.3.2 Entropy regularization

1703 In this case, entropy is useful to ensure that the reliability maps are not too
 1704 homogeneous. The entropy regularization term encourages the model to assign
 1705 more discriminative reliability values by penalizing uniform distributions. This
 1706 prevents the model from converging to uninformative reliability estimates where
 1707 all labelers appear equally reliable, thus maintaining meaningful distinctions
 1708 between more and less trustworthy annotators.

$$\mathcal{L}_{entropy} = \alpha_{entropy} \cdot \mathbb{E}_{r \in R} [\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))] \quad (5-2)$$

1709 The behavior of this regularization term is shown in Figure 5-3 (b).

1710 5.3.3 Sum-to-one regularization

1711 The reliability branch for scalar, pixel and features approaches, might or might not
 1712 be forced to sum to one depending on the output activation function ². For the
 1713 cases in which this sum is not forced to be unitary (sigmoid) the flexibility of this
 1714 sum is manageable through a regularization term which mildly penalizes the model
 1715 for adding to values too far from the unit. This was found useful in situations in
 1716 which the reliability maps were not expected to be punctuated (e.g. for the scalar
 1717 reliability) as $1/R$, but rather close to the unit for all of them (if all labelers were
 1718 good). This regularization term was found particularly useful in situations where
 1719 labelers have good and similar levels of expertise. ³

²This was left as flexible and configurable in the Network Architecture Building API from `seg_tgce`, allowing it to be either softmax or sigmoid.

³The behavior of this regularization term of focusing in the homogeneity of the reliability maps is mostly covered by the combination of the centroid and entropy regularization terms, which was proven to be redundant in section 5.4.1.

$$\mathcal{L}_{sum} = \alpha_{sum} \cdot \left(\sum_{\forall r \in R} \Lambda_r - 1 \right)^2 \quad (5-3)$$

1720 The behavior of this regularization term is shown in Figure 5-3 (c).

1721 5.3.4 Missing labels penalty

1722 In situations where the datasets have a low labeling rate, the model might take an
 1723 optimization shortcut by assigning high reliability values to labelers which did not
 1724 label at all, since the element-wise product with the mask would be close to zero,
 1725 which favors a minimization target. To avoid this, a term was added to the loss
 1726 function which penalizes the model for assigning high reliability values to unlabeled
 1727 patches, as follows:

$$\mathcal{L}_{\lambda} = \alpha_{\lambda} \cdot \max \left(0, \sum_{\forall r \in R} (\Lambda_r - \mu_r) \cdot e^{\frac{\Lambda_r - \mu_r - 1}{\beta}} \right) \quad (5-4)$$

1728 Where Λ_r is the estimated reliability of the labeler r from the reliability branch for
 1729 a certain patch, μ_r is the encoded presence of the label mask for labeler r for the
 1730 same patch, β is a hyperparameter which control the penalty strength for high and
 1731 low estimated reliabilities, respectively.

1732 The behavior of this regularization term is shown in Figure 5-3 (d).

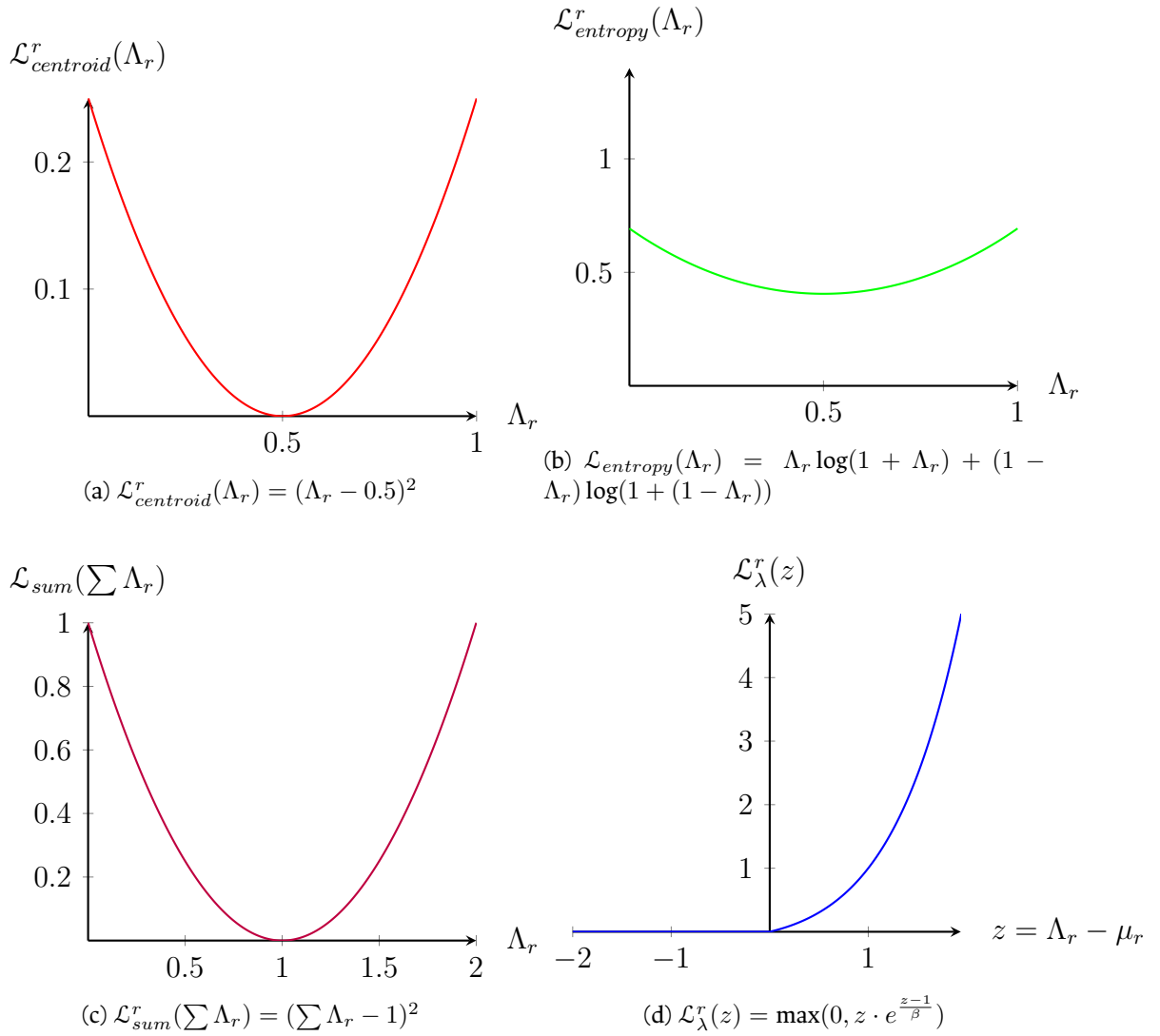


Figure 5-3 Visualization of the four regularization terms used in the TGCE loss function: (a) centroid penalty for extreme reliabilities, (b) entropy regularization, (c) sum-to-one regularization, and (d) missing labels penalty. Each subplot shows how the respective regularization term behaves as a function of its input variables.

1733 This function applies no penalty for cases in which $\Lambda_r < \mu_r$, meaning no penalty
 1734 for assigning any reliability value (ranging from 0 to 1) to a labeler who actually
 1735 labeled a given patch. On the other hand, a penalty is indeed applied in cases where
 1736 $\Lambda_r > \mu_r$, meaning a penalty for assigning high reliability values to a labeler which

1737 did not label at all. The proportion on how high and how quickly (in the space of
 1738 difference $\Lambda_r - \mu_r$) the penalty is applied is controlled by the hyperparameters α_λ
 1739 and β . The behavior of this term is shown in Figure 5-3 (d).

1740 These regularization terms lead to the overall loss function being in the form:

$$\begin{aligned}
 L_{TGCE_{reg}} = L_{TGCE_{ss}} & \quad (5-5) \\
 & + \alpha_{reg} \cdot \mathbb{E}_r[(\Lambda_r - 0.5)^2] \\
 & + \alpha_{entropy} \cdot \mathbb{E}_r[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))] \\
 & + \alpha_{sum} \cdot \left(\sum_{\forall r \in R} \Lambda_r - 1 \right)^2 \\
 & + \alpha_\lambda \cdot \max \left(0, \sum_{\forall r \in R} (\Lambda_r - \mu_r) \cdot e^{\frac{\Lambda_r - \mu_r - 1}{\beta}} \right)
 \end{aligned}$$

1741 For the scalar reliability approach, the loss function operates on global reliability
 1742 values per annotator. The reliability maps are expanded to match the spatial
 1743 dimensions of the predictions through tiling operations. This approach is
 1744 computationally efficient but provides the same reliability value across all spatial
 1745 locations.

1746 The feature-based reliability approach requires additional processing to handle the
 1747 spatial dimensions of the bottleneck features. The reliability maps are resized
 1748 using bilinear interpolation to match the spatial dimensions of the predictions.
 1749 This approach provides coarse-grained spatial reliability information while
 1750 maintaining computational efficiency.

1751 The pixel-wise reliability approach operates directly on full-resolution reliability
 1752 maps, requiring no spatial transformations. This approach provides the most
 1753 detailed spatial reliability information but requires more computational resources
 1754 and is more prone to overfitting in sparse datasets (low labeling rate).

5.4 Pretraining Process

The training process begins with the initialization of the ResNet-34 backbone using pre-trained weights, providing a strong foundation for feature extraction. The model's hyperparameters are optimized using Optuna [Akiba et al., 2019], a hyperparameter optimization framework that employs Bayesian optimization to search for optimal values of:

- Learning rate: Sampled logarithmically between 10^{-6} and 10^{-2}
- TGCE_{SS} parameters:
 - q : Controls truncation of the loss function (0.1 to 0.9)
 - Noise tolerance: Controls model's robustness to label noise (0.1 to 0.9)
 - Regularization weights: Control the balance between different loss terms.

5.4.1 Hyperparameter Optimization Pretraining Study

A hyperparameter optimization study was conducted for every single reliability branch estimation type (scalar, pixel and features) using Optuna, each with 20 iterations, on the Oxford PET dataset with three scorers emulating SNR noises of -20, 0, and 10 dB respectively, and a labeling rate of 0.5.

The analysis of hyperparameter importance reveals crucial insights about the regularization terms' contributions to model performance. As shown in Figure 5-4, the contour plots visualize the optimization landscape for different regularization parameters, providing complementary evidence to the importance scores. Table 5-1 presents the hyperparameter importance ranking for the regularizers study for each reliability estimation approach.

The contour visualizations (Figure 5-4) and importance analysis consistently demonstrate that the sum regularization (α_{sum}) exhibits the lowest importance across all reliability estimation approaches, with scores of 0.0756, 0.0661, and 0.0219 for pixel, scalar, and features approaches respectively. This low importance, combined with the relatively flat contour regions observed in the optimization landscape, suggests that the sum regularization's contribution might be largely redundant with the combination of centroid and entropy regularization terms. This finding has significant implications for model design: since sum regularization requires sigmoid activation in the output layer, while centroid and entropy regularization can work with softmax outputs, eliminating the sum regularization term allows for the use of fully softmax outputs in subsequent experiments. This simplification reduces the hyperparameter search space and potentially improves training stability, making the optimization process more efficient for future experiments. This is likely to happen because of the combination of entropy and centroid regularization terms, which are also related to controlling homogeneity or extreme reliability values, leading to an undermining of the sum regularization term, in such scenario, the sum regularization term would be redundant.

Hyperparameter	Importance Score		
	Pixel	Scalar	Features
α_{sum}	0.0756	0.0661	0.0219
α_{entropy}	0.0268	0.1342	0.5395
α_{λ}	0.0234	0.3362	0.0282
β	0.0200	0.2999	0.0218
a	0.1738	0.1636	0.3886

Table 5-1 Hyperparameter importance ranking for regularization parameters obtained from Optuna optimization study after 20 iterations for each reliability estimation approach (pixel, scalar, and features). This study used preselected values of $q = 0.66$, $\tilde{C} = 0.62$, and learning rate $= 6.5 \times 10^{-4}$. The importance scores indicate the relative impact of each regularization hyperparameter on the model's performance for each specific approach.

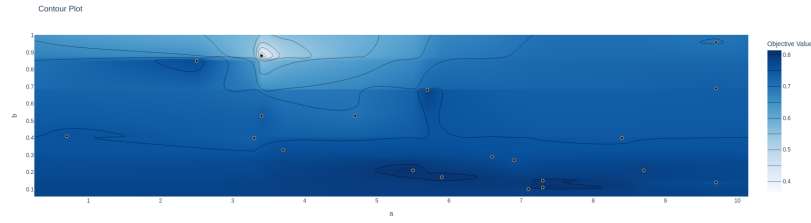
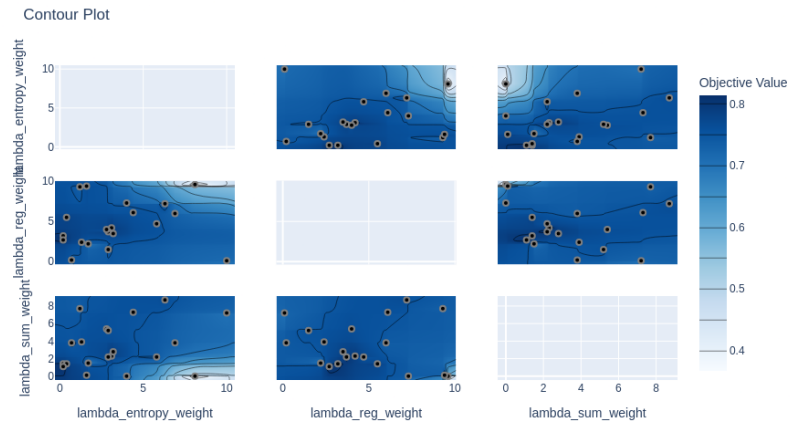
(a) Missing labels penalty regularization (α_λ and β).(b) Other regularization terms (α_{centroid} , α_{sum} , α_{entropy}).

Figure 5-4 Contours of the tuner optimization study for the Oxford PET dataset with the best hyperparameters found for the comprehensive hyperparameter optimization study.

5.4.2 Training Process Overview

After finding the optimal hyperparameters, the model is trained end-to-end using the Adam optimizer with the best learning rate found⁴. The TGCE_{SS} loss function plays a crucial role in updating both the segmentation and reliability branches, ensuring that the model learns to effectively handle multiple annotators' inputs while accounting for their varying reliability.

The training process illustrated in Figure 5-5 demonstrates the comprehensive approach adopted for model optimization. This workflow integrates both the

⁴For every experiment this was also tuned with the help of the Optuna study, having different optimal values for every experiment.

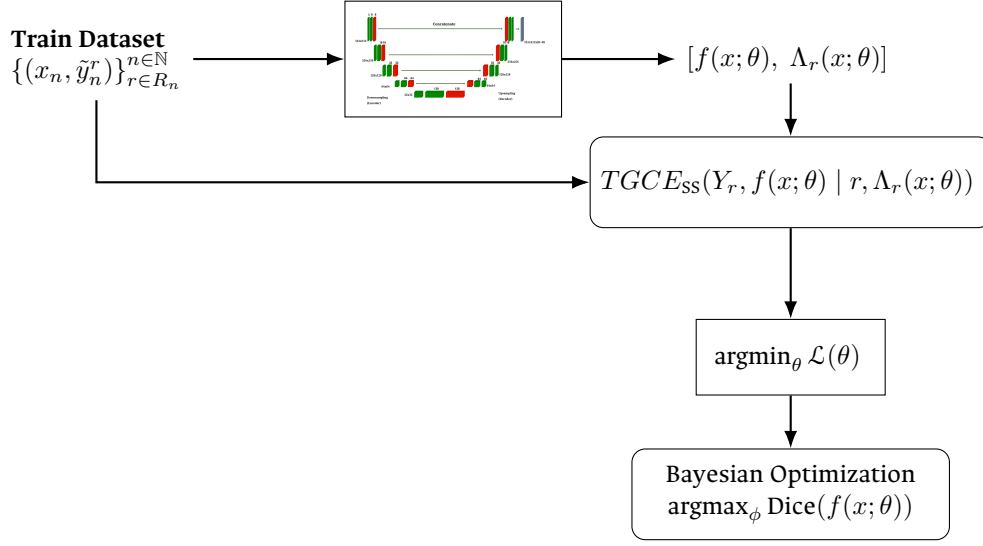


Figure 5-5 Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters ϕ tuning are based on the Dice coefficient of the validation data set (with maximization target).

1804 multi-annotator dataset structure and the dual-objective optimization strategy,
 1805 where the model parameters θ are optimized to minimize the $TGCE_{SS}$ loss
 1806 function while hyperparameters ϕ are tuned to maximize the Dice coefficient on
 1807 validation data. The visual representation emphasizes the iterative nature of the
 1808 training process, highlighting how the reliability estimation branch $\Lambda_r(x; \theta)$ works
 1809 in conjunction with the segmentation branch $f(x; \theta)$ to provide robust predictions
 1810 that account for annotator variability. This systematic approach ensures that the
 1811 model not only learns to perform accurate segmentation but also develops an
 1812 understanding of the reliability patterns inherent in multi-annotator medical
 1813 imaging datasets.

5.5 Experimental Setup

5.5.1 Noisy Annotations Generation

A challenge arises for the creation of emulated noisy annotations datasets, since it is expected for images annotations to have some degree of expertise variability, similar to what is expected in real crowdsourced datasets. Simply introducing random noise into the annotated masks does not work, since the original morphological structures from the expected ground truth are far from being preserved. Instead, a “noisy” labeler is expected to produce an annotation which has at least some degree of morphological consistency on itself, even if it shows discrepancies when compared with some metric (like DICE score) against the ground truth annotation.

This has been proven experimentally when introducing random noise into any popular segmentation dataset, in which the resulting mask is just a non coherent map of noise across the image, as shown in Figure 5-6.

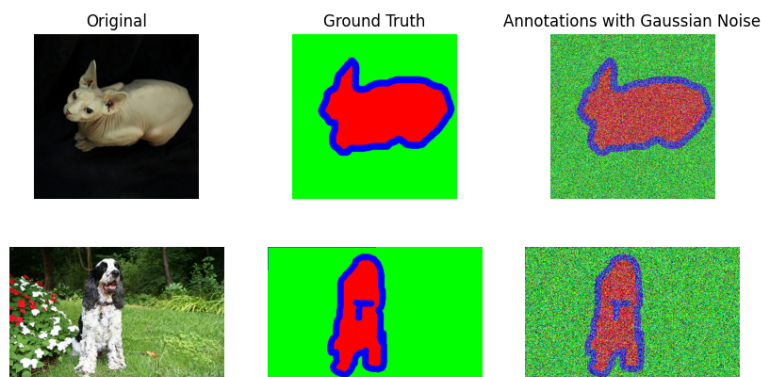


Figure 5-6 Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

For this reason, a more sophisticated approach was needed to create datasets with emulated noisy annotations. The approach used here was to use a pre-trained

U-Net model to produce a good enough ⁵ segmentation of the image, which was then used to produce a “noisy” annotation by introducing random noise into the last encoder layers of the model, thus preserving the morphological consistency of the original annotation. This strategy works since slight modifications introduced into the encoder layers weights, resemble conceptual disturbances with respect to the original ground truth label, which kind of emulates the cultural behavior of having a different interpretation of the image, either for having a different level of expertise or for having a different point of view or school of thought. The first encoding layers were not modified, since it is expected human labelers would agree on the most fundamental structures (analog to extracted features from initial convolutional layers) in a similar way, thus preserving some degree of morphological consistency of the original annotation.

In this way, the level of “disturbance” with respect to the ground truth for an emulated annotator can be controlled by the level of noise introduced into the weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \quad (5-6)$$

where $\mathbf{W}_{original}$ represents the original weights of the last encoder layers, ϵ_d follows a Gaussian distribution with zero mean and variance σ^2 . The variance σ^2 controls the level of noise introduced, and thus the degree of disturbance in the resulting segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure 5-7. It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

⁵More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

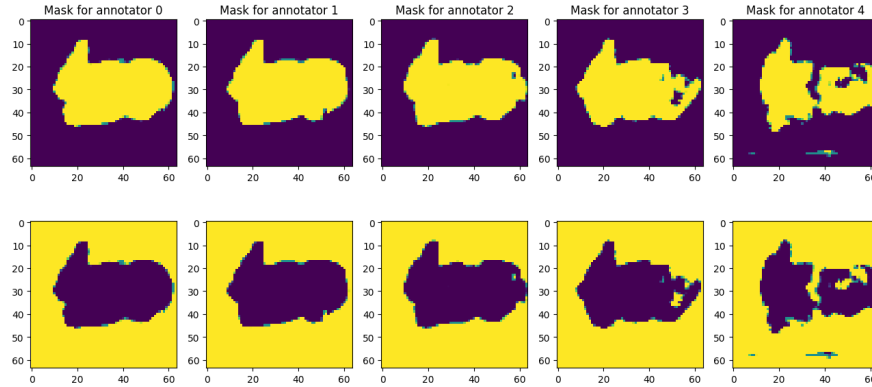


Figure 5-7 Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

5.5.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate $r_l \in [0, 1]$. The labeling rate r_l controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in experiments with variable amount of labelers, noise levels, and labeling rates. An overview of the algorithm is shown in Algorithm 5-1. A sample of the generated noisy and missing labels dataset is shown in Figure 5-8.

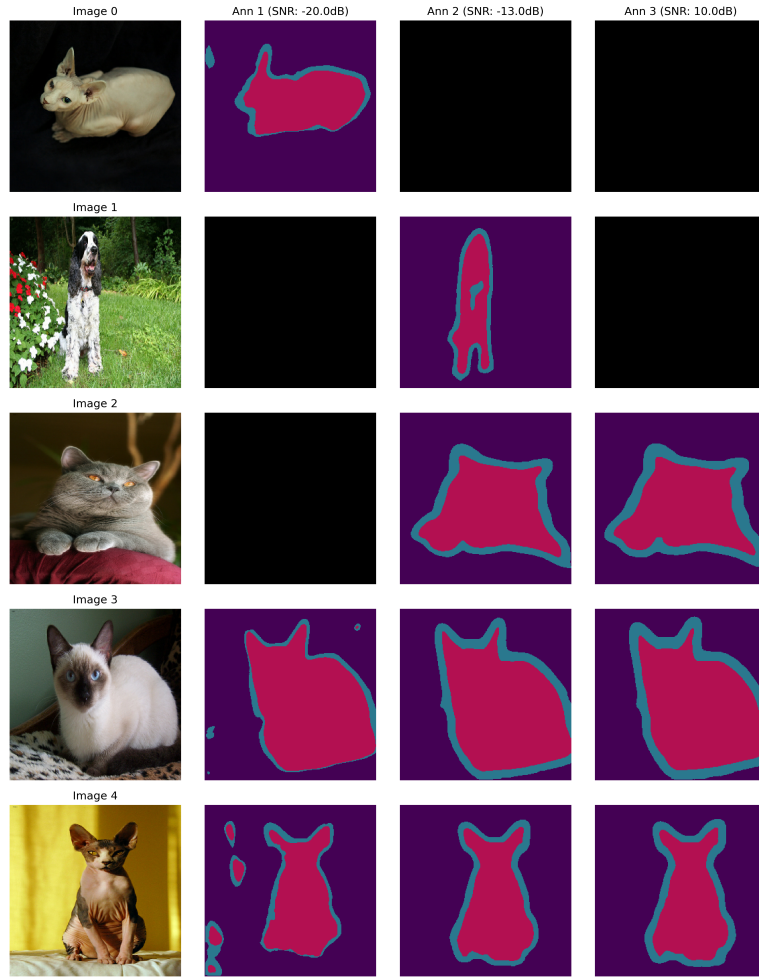


Figure 5-8 Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

1867 5.6 Results

1868 Initially, a baseline model (ground truth only) was trained with an early stopping
 1869 mechanism, which stopped the training when the validation loss stopped decreasing
 1870 for 10 epochs, reaching only 20 epochs of significant training. The results are shown
 1871 in Figure 5-9.

Algorithm 5-1 Multi-Annotator Dataset Generation with Noisy and Missing Labels

Require: Base segmentation model M , SNR levels $\{s_1, \dots, s_n\}$, labeling rate $r \in [0, 1]$

Ensure: Dataset with multiple noisy annotations and missing labels

```

1: Step 1: Generate Noisy Annotators
2: for each SNR level  $s_i$  do
3:   Create disturbed model  $M_i$  by adding noise to  $M$  with SNR  $s_i$ 
4: end for
5: Step 2: Initialize Labeler Assignment System
6: Create assignment matrix  $A \in \{0, 1\}^{N \times n}$  where  $N$  is number of samples
7: for each annotator  $j \in \{1, \dots, n\}$  do
8:   for each sample  $i \in \{1, \dots, N\}$  do
9:      $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$ 
10:   end for
11: end for
12: Step 3: Generate Multi-Annotator Dataset
13: for each image  $x$  in dataset do
14:   Resize  $x$  to target dimensions
15:   for each annotator model  $M_i$  do
16:     Generate annotation  $y_i = M_i(x)$ 
17:     Apply labeler assignment mask:  $y_i \leftarrow y_i \cdot A_{i,j}$ 
18:   end for
19:   Store  $(x, \{y_1, \dots, y_n\})$  in dataset
20: end for
   return Multi-annotator dataset with noisy and missing labels

```

Then, using the scalar model described in Section 5.2.1, training was performed for 20 epochs and Bayesian hyperparameter optimization on learning rate, decoder dropout rate, for the regularizers described in Section 5.3. The results are shown in Figure 5-10.

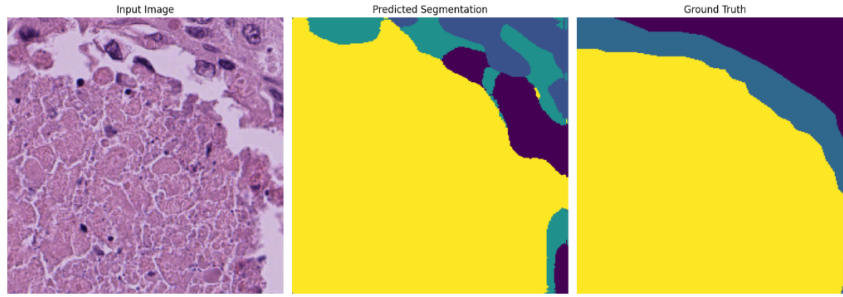


Figure 5-9 Segmentation results for baseline model on the Triple-Negative Breast Cancer (TNBC) (Crowd-Seg) dataset after 20 epochs of training. Even though the segmentation algorithm seems to show progress on its way to capture the underlying target structures to be segmented, since the ground truth annotation is not expressive enough, the Dice coefficient remains below 0.55, which is not considered a good performance. The baseline model remained with overfitting even after combining techniques of: Bayesian hyperparameter optimization, data augmentation, learning rate scheduling, dataset shuffling, and early stopping.

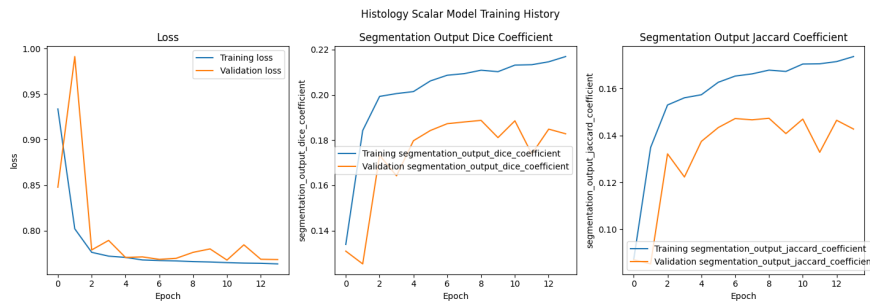


Figure 5-10 Segmentation results for scalar model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Again, training metrics show a constant growth, but not validation metrics, which shows a combined issue of overfitting and unreliable ground truth.

5.6.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8718 and 0.7917 respectively) for the dataset with no missing labels (label rate of 1.0), our proposed TGCE_{SS} approaches show superior performance for label rates below 1.0 (Table 5-2). The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information. It is of note that the key strength of TGCE_{SS} lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

Method	Label Rate 1.0		Label Rate 0.7		Label Rate 0.3	
	Dice	Jaccard	Dice	Jaccard	Dice	Jaccard
TGCE scalar	0.8622±0.009	0.7791±0.012	0.8568±0.017	0.7726±0.024	0.8529±0.004	0.7677±0.007
TGCE feature	0.6995±0.044	0.5733±0.047	0.4766±0.026	0.3441±0.027	0.4766±0.026	0.3441±0.027
TGCE pixel	0.8468±0.018	0.7589±0.025	0.8545±0.013	0.7695±0.017	0.8283±0.001	0.7359±0.002
MV	0.8132±0	0.7222±0	0.7572±0	0.6554±0	0.5566±0	0.4687±0
STAPLE	0.8718±0	0.7917±0	0.8439±0	0.7609±0	0.5734±0	0.5158±0

Table 5-2 Performance metrics (Dice and Jaccard) at different labeling rates. Best values per category are bolded.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures 5-11 and 5-12. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator reliability is a significant advantage over traditional approaches like STAPLE and MV, which focus solely on segmentation accuracy without providing insights into annotator behavior.

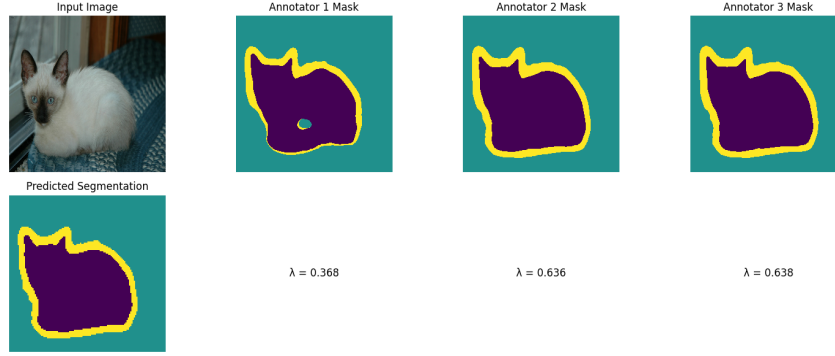


Figure 5-11 Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$.

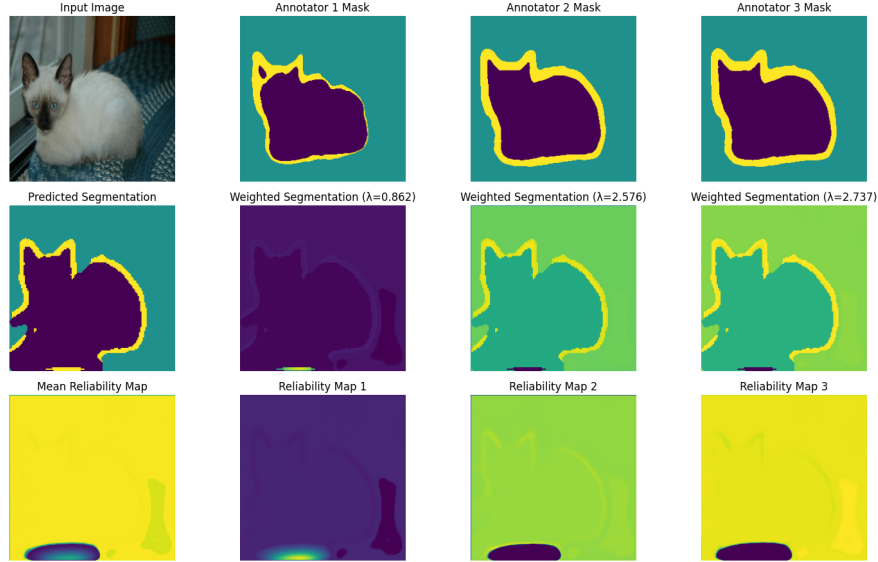


Figure 5-12 Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$.

5.6.2 TNBC Dataset

As mentioned in Section 2.4.2, [Amgad et al., 2019] presented a TNBC dataset, which is crowdsourced and available as HistomicsTK metadata. Later, [López-Pérez et al., 2023] sampled 151 Whole Slide Imagings (WSIs), from which 161 Region of Interests (ROIs) were extracted and labeled by 20 medical students. The dataset was sampled as patches of size 512x512, resulting in 10173 training patches, 1264 validation patches and 399 testing patches. The segmentation masks for the patches are composed of the classes: Tumor, Stroma, Inflammation, Necrosis, and Others⁶. This subset of the TNBC is named Crowd-Seg [López-Pérez et al., 2023].

Raw PNG files for the images patches and masks were provided by the authors, in practice, in order to optimize the use of these data, a custom tensorflow dataset was built for facilitating batch memory, caching and prefetching. The custom dataset implementation handles the loading and preprocessing of both images and masks, including normalization of images to the [0,1] range and one-hot encoding of the segmentation masks. It also manages the multi-annotator structure by creating a labeler mask that indicates which annotators provided labels for each sample. The dataset supports data augmentation through random transformations (flips, rotations, brightness/contrast adjustments) and includes visualization utilities for debugging and analysis purposes. This custom dataset implementation is publicly available^{7 8} within the PyPI package included in the overall solution of this work. (see Section 6.3).

Notes on the Crowd-Seg Dataset

The multiple labelers version of the TNBC dataset (Crowd-Seg) contains a couple of peculiarities that are worth mentioning. First, the label rate (inverse percentage of

⁶In practice, some of this masks also contain an additional class labeled as 'ignore'.

⁷https://github.com/blotero/seg_tgce

⁸<https://seg-tgce.readthedocs.io/en/latest/>

labels that are missing) is not uniform across the labelers and, in general, quite low, ranging from 1 to 12% as seen in Figure 5-13.

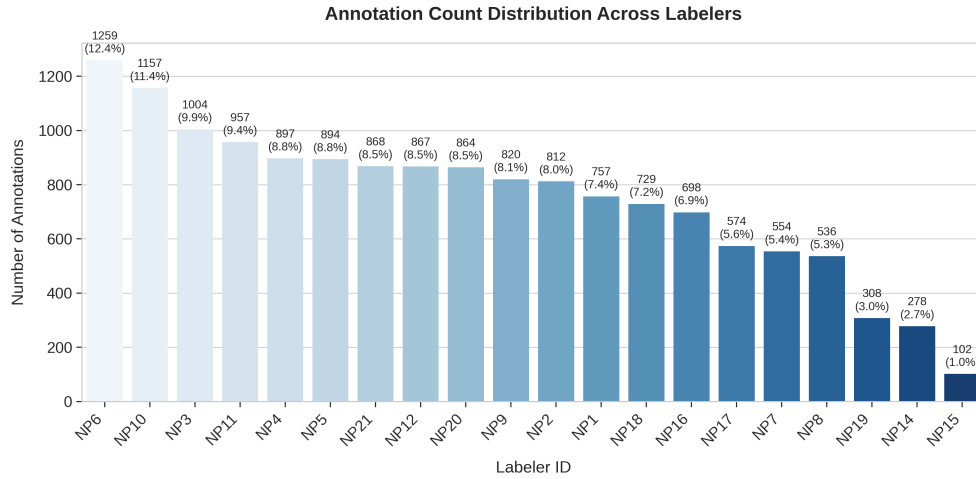


Figure 5-13 Label rate distribution across the labelers for the TNBC (Crowd-Seg) dataset.

On the other hand, the classes distribution across all patches are not equal, being ‘Tumor’ (class 2) the most frequent class. Despite the classes not being equally distributed, the distribution of the classes across the labelers is fairly uniform, as shown in Figure 5-14.

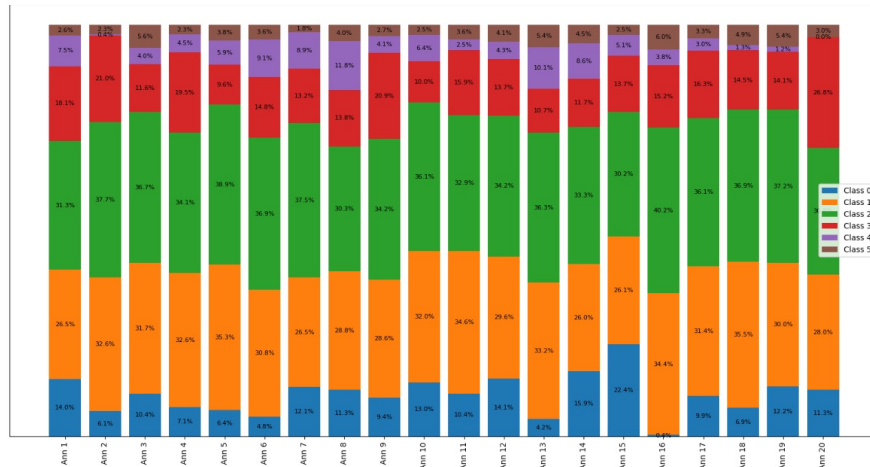


Figure 5-14 Classes distribution across the labelers for the TNBC (Crowd-Seg) dataset. Most common classes are ‘Tumor’ (class 2), ‘Other’ (class 1) and ‘Stroma’ (class 3).

It was also found that the dice score of the labelers masks against the ground truth differed across the labelers and the classes, as shown in Figure 5-15. The target class (Tumor) ranged dice scores across the labelers from 0.5 to 0.9, other classes can get as low as 0.4 for in average for some labelers, meaning very high levels of disagreement.

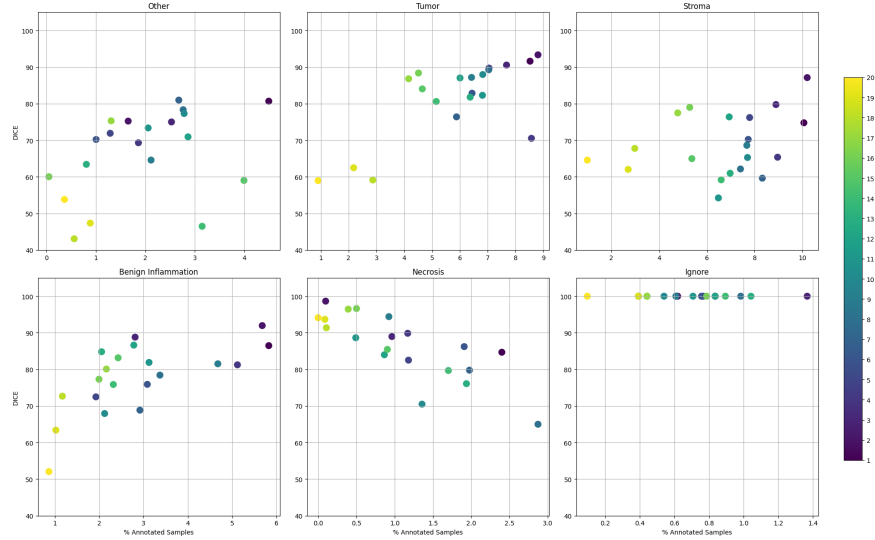


Figure 5-15 Dice score of the labelers masks against the ground truth for the TNBC (CrowdSeg) dataset.

Finally, it was found that the expert labels, which would thereby be considered as ground truth for metrics evaluation, were not always correct, as shown in Figure 5-16. These morphological inconsistencies make an objective metric for evaluating the performance of the proposed model difficult to define, since the ground truth is not always correct or expressive enough.

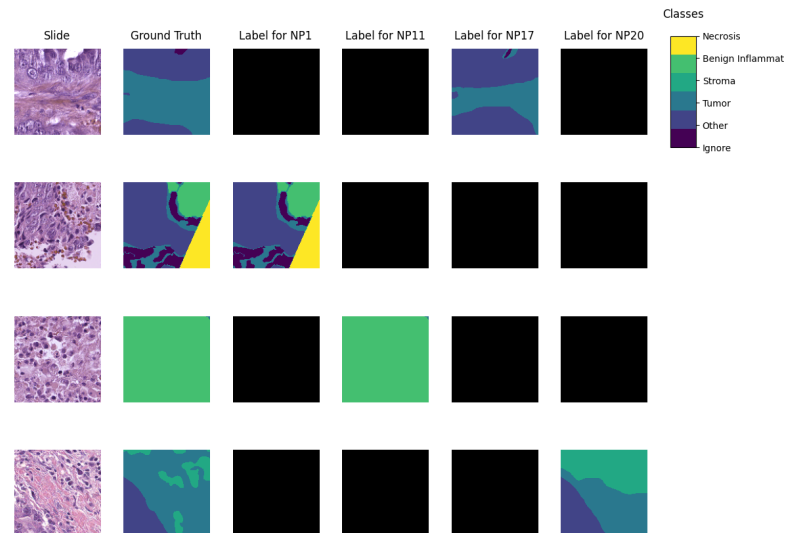


Figure 5-16 Example of ground truth with odd geometries (second patch from top to bottom), which are not always correct or compatible with the underlying structures of the patch itself.



Figure 5-17 Training curves for the baseline model on the TNBC (Crowd-Seg) dataset.

5.7 Summary

The experimental results on both the Oxford-IIIT Pet and TNBC (Crowd-Seg) datasets provide valuable insights into the performance and limitations of our

proposed approach. The model demonstrated exceptional performance on the Oxford-IIIT Pet dataset, achieving competitive Dice and Jaccard scores while successfully capturing annotator reliability patterns. The model's ability to simultaneously learn segmentation and reliability estimation proved particularly valuable, providing insights into annotator behavior that traditional approaches like STAPLE and MV cannot offer.

However, the experiments on the **TNBC** dataset revealed significant challenges in evaluating semantic segmentation models for histopathological images when dealing with unreliable ground truth. The morphological inconsistencies in the expert annotations, combined with the extremely low label rates (1-12%) across annotators, made it difficult to objectively measure the model's performance. While the model showed promising convergence in training, the validation metrics remained unstable, suggesting that the current dataset may not be suitable for rigorous evaluation of multi-annotator segmentation approaches.

These findings highlight the need for more reliable ground truth annotations in histopathological image segmentation datasets. Future work should focus on:

- Developing datasets with more consistent and reliable expert annotations
- Implementing stricter quality control measures for ground truth generation
- Exploring alternative evaluation metrics that account for annotation uncertainty
- Investigating methods to handle extremely sparse annotations more effectively

Despite these challenges, the model's performance on the Oxford-IIIT Pet dataset demonstrates its effectiveness in handling multiple annotators with varying reliability levels. The model successfully learned to:

- Balance the contributions of different annotators based on their reliability

- 1965 • Provide detailed spatial reliability information through pixel-wise estimation
- 1966 • Maintain high segmentation accuracy even with missing labels
- 1967 • Adapt to different noise levels in annotator inputs

1968 These capabilities make the proposed approach particularly valuable for
1969 real-world applications where multiple annotators with varying expertise levels
1970 contribute to the dataset. The model’s ability to simultaneously learn
1971 segmentation and reliability estimation provides a more comprehensive
1972 understanding of the annotation process, which could be leveraged to improve the
1973 quality of future annotations and dataset construction.

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CHAPTER

SIX

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FINAL REMARKS

1979

6.1 Conclusions

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This thesis has presented a novel approach to address the challenges of Image Semantic Segmentation (ISS) in histopathology images, particularly in crowdsourcing scenarios where multiple annotators with varying levels of expertise contribute to the labeling process. The proposed solution successfully tackles the fundamental challenges of modeling annotator performance and handling label inconsistencies while maintaining computational efficiency. The work demonstrates significant improvements over existing state-of-the-art methods, particularly in scenarios with noisy and missing labels, which are common in crowdsourcing approaches.

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Throughout this work, several techniques have been explored to develop a model that can face Image Semantic Segmentation (ISS) of real world histopathology images in crowdsourcing scenarios. Different approaches were studied and evaluated to reach an optimal solution which could model the annotators' performance from the input space. The following remarks are based on the results obtained from the experiments conducted and reviewed literature:

- 1995 • Chained Gaussian Processes approaches, while robust for classification and
1996 regression tasks, proved computationally intensive for image segmentation.
1997 This limitation highlighted the need for more efficient approaches that could
1998 handle the complexity of medical image segmentation while modeling
1999 annotator performance.
- 2000 • The development of a novel loss function was crucial for the success of this
2001 work, as existing loss functions were insufficient to model the annotators'
2002 performance across the input space. The proposed loss function successfully
2003 enabled the model to adaptively infer the best possible segmentation without
2004 requiring separate inputs about labeler performance.
- 2005 • The introduction of a tensor map to codify labeler reliability proved to be an
2006 effective approach, allowing the model to implicitly weigh the labelers based
2007 on their performance across the mask and classes space, thus handling the
2008 labeler performance inconsistency effectively.
- 2009 • The use of U-Shaped Deep Learning models as a building block for the
2010 proposed solution was validated, as it effectively captured spatial
2011 relationships between pixels and annotations while handling the variability
2012 in image quality.
- 2013 • The combination of Chained Deep Learning approaches with the proposed
2014 loss function and tensor map representation resulted in a comprehensive
2015 solution being, to the best of our knowledge, the first approach on offering
2016 an image segmentation solution for multiple labelers in medical image with
2017 interpretability in the reliability of the labelers, successfully addressing both
2018 labeler inconsistency and image quality variability. The model outperformed
2019 other state of the art solutions in terms of segmentation accuracy in noisy
2020 and missing labels conditions, which is rather common in crowdsourcing
2021 approaches.

6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes¹ evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of **Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA)** as a building block for the proposed solution could be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of **CNNs** to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging:** The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency:** For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability:** The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.

¹Including GPFlow, and GPFlux.

2046 • **Pathology-specific Challenges:** The emulation could be tailored to simulate
2047 expert disagreements in specific pathological conditions, where the
2048 morphological features might be more ambiguous or subject to
2049 interpretation.

2050 These extensions would help validate the model's performance across a broader
2051 range of medical imaging scenarios and ensure its robustness in real-world clinical
2052 applications.

2053 The pixel-wise reliability mapping approach shows particular promise for future
2054 development. While the current implementation already demonstrates good
2055 performance, several enhancements could lead to significant improvements:

2056 • **Advanced Data Augmentation:** Implementing a comprehensive data
2057 augmentation strategy could significantly improve the model's convergence
2058 and performance. This could include:

- 2059 – Geometric transformations (rotations, flips, elastic deformations)
- 2060 – Intensity variations to simulate different staining conditions
- 2061 – Multi-scale training to handle varying tissue sizes
- 2062 – Adversarial training to improve robustness

2063 These augmentations would help the model learn more robust features and
2064 better generalize to unseen data.

2065 • **Attention Mechanisms:** Integrating attention mechanisms into the pixel-wise
2066 reliability branch could help the model focus on the most relevant regions for
2067 reliability estimation, potentially improving both accuracy and computational
2068 efficiency.

2069 • **Multi-scale Reliability Estimation:** Implementing a hierarchical approach to
2070 reliability estimation could capture both fine-grained local variations and
2071 broader regional patterns in annotator reliability, this would imply a
2072 network with yet more branches and outputs.

The combination of these enhancements with the existing pixel-wise reliability mapping approach could potentially outperform all other techniques, as it would provide the most detailed and accurate representation of annotator reliability while maintaining computational efficiency. The use of data augmentation, in particular, could help the model converge faster and achieve better generalization, as it would expose the model to a wider variety of possible variations in both the input images and the corresponding annotations.

6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository² has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the $TGCE_{SS}$ loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. The two `tfds` datasets available in the repository are:

- **Triple Negative Breast Cancer (TNBC) Dataset:** A dataset of histopathology images of breast cancer, with annotations of the tumor regions from [López-Pérez et al., 2024]. The dataset is automatically fetched from AWS S3, which eliminates the need for manual downloads or external dependencies.
- **Oxford-IIIT Pet Dataset:** A dataset of pet images, based on the well-known Oxford-IIIT Pet Dataset with annotations of the pet regions. The mechanism for emulating noisy crowdsourced annotations exposed throughout the present work (see Section 5.5.1) is also available in the repository in a tunable manner, just as the TNBC dataset, this `tfds` is batteries included

²https://github.com/blotero/seg_tgce

2098 (automatic fetching and mapping, no need to manually download any
2099 external source).

2100 To further simplify the adoption of the proposed solution, a PyPI package has been
2101 published with detailed documentation available at
2102 <https://seg-tgce.readthedocs.io/en/latest/>. The documentation provides
2103 comprehensive installation instructions, API reference, usage examples, and best
2104 practices for implementation. This package makes it straightforward for the
2105 community to integrate the proposed approach into their existing workflows, with
2106 a focus on ease of use and clear documentation of all available features and
2107 options.

2108 An academic publication is currently being prepared for submission to a Q1
2109 journal. This publication will provide detailed theoretical foundations of the
2110 proposed approach, comprehensive experimental results and analysis, comparison
2111 with state-of-the-art methods, and discussion of limitations and future directions.
2112 The publication will serve as the primary reference for the theoretical and practical
2113 aspects of the proposed solution, ensuring that the work is properly documented
2114 and validated through the peer-review process.

2115 These resources collectively contribute to the reproducibility and accessibility of
2116 the research, enabling the broader scientific community to build upon and extend
2117 the proposed approach. The combination of open-source code, well-documented
2118 APIs, and peer-reviewed publication ensures that the work can be properly
2119 evaluated, validated, and utilized by other researchers in the field. This
2120 comprehensive approach to sharing research outputs aligns with current best
2121 practices in scientific computing and Machine Learning research, promoting both
2122 transparency and practical utility of the proposed solution.

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